General examination in pediatrics

Differences in Pediatric Physical Examination Compared to Adults

- Gather as much data as possible by observation first
- Position of child: parent's lap vs exam table
- Stay at the child's level as much as possible.
- Order of exam: least distressing to most distressing.
- Distraction is a valuable tool
- Examine painful area last

Approach to examining children

Obtaining child's cooperation

- Explain what you are about to do,
- and what you want the child to do,
- Examination is essential . Don't ask his permission!

...LOL

- Smile and talk,
- Leave unpleasant procedures last.



Approach to examining children

Adapting to child's age

- Babies in first months best examined on examination couch with parent next to them,
- A toddler is best initially examined on mothers lap ,
- or over parents shoulder ,
- Preschool children may be examined while playing,
- Older children and teenagers, concerned about privacy.
- Teenager girls in presence of mother, nurse or chaperone. Be aware of sensitivities in ethnic groups.



Approach to examining children

Warm clean hands

- Hands must be washed before and after examining the child,
- "Warm smile, warm hands",
- ..warm stethoscope also helps.



Pediatric age groups:

Neonate: 0 day – 28 day

Infant: 29 day **-** 1year

Toddler: 1-3 years

Children: 3– 12years

Adolescent: 12 yr-18yr

General examination:

General look: (well, ill, toxic)

- **position in bed** : comfortable or not
- conscious level
- **Cardinal signs** : for pallor/jaundice/ cyanosis
- **Hydration status** : well hydrated , dehydrated : sunken eye , sunken fontanel, dry tongue , no tears. over hydrated :puffy eyelids
- **body built**: average, under or over built,
- **Vital signs:** pulse, temperature, respiratory rate, blood pressure
- ***anthropometric measurements :***

weight ,length\height, head circumference.

Put them on growth charts according to age.

- gait / abnormal facial features / apparent congenital malformation/ cubitus.

Vital signs

- temperature
- heart rate
- respiratory rate
- blood pressure

Normally child differ than adults, and vary according to age

Temperature:

- Grade:

 average normal 37 ˚C (98.6 ˚F)

 Infant & children > 1 years old :

normal value 36.6(97.9 ˚F) -37.2 ˚C(99 ˚F)

 up to 37.7 ˚C for <1 years old

- Route: oral(same reading) vs. Axillary (+0.5˚C) vs. rectal (-0.5˚C)
- Fever> 37.7˚C

Low grade < 38.5 ˚C

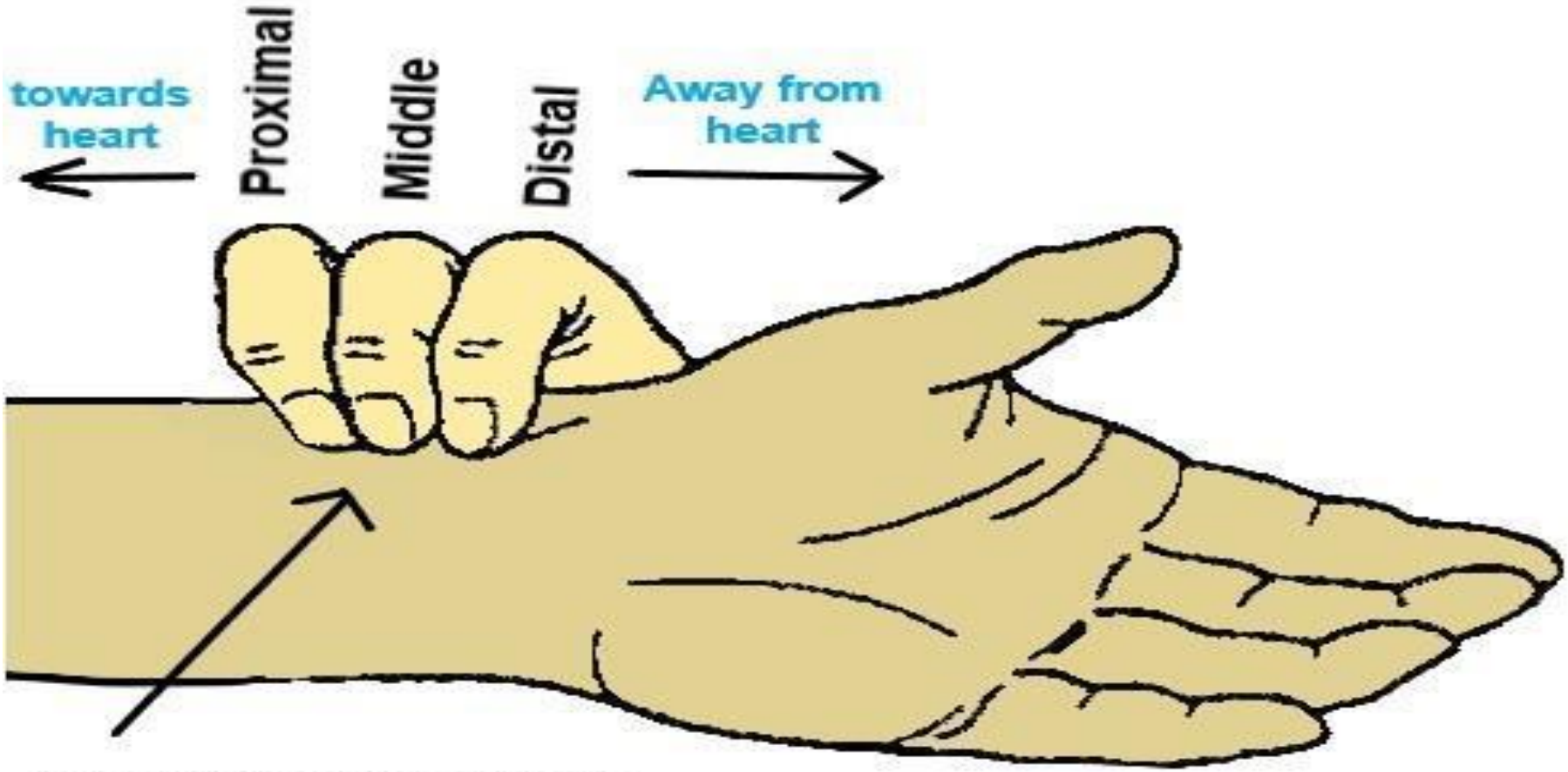
High grade (38.5 –41 ˚C)

Hyperpyrexia > 41 ˚C

Heart rate:

- Comment on rate, rhythm ,volume , special character.
- Rate (newborn 140 – infant 120 – preschool 100 – school children 90) ±20 beat / min.
- Beat / minute according to age.
- Auscultate or palpate apical pulse or palpate femoral pulse in infant.

Palpate ant. cubital or radial pulse in older child.



keep fingers in a line against each other without gap.

Palpation of radial artery:



Palpation of dorsalis pedis artery:



Palpation of brachial artery:



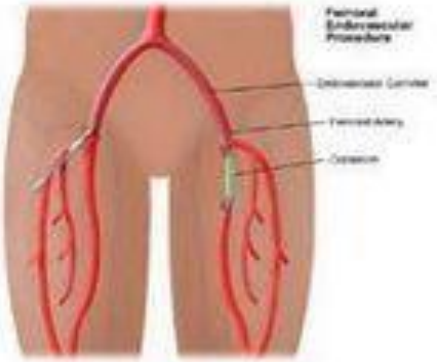
Palpation of arterial pulses



Radial



Brachial



Femoral



. Popliteal



Dorsalis pedis



Posterior tibial

respiratory rate:

Observe for a minute. Infants normally have periodic breathing so that observing for only 15 seconds will result in a skewed number.

- **Blood pressure** :

Appropriate size cuff 2/3width of upper arm.

The **WRONG SIZE CUFF** can give you the **WRONG READINGS**

If Cuff Too SMALL



High Reading

If Cuff Too LARGE



Low Reading

Measure your arm for the correct fit

SELECTING THE CORRECT BLOOD PRESSURE CUFF

- Width of the inflatable bladder of the cuff should be about 40% of upper arm circumference (about 12–14 cm in the average adult)
- Length of inflatable bladder should be about 80% of upper arm circumference (almost long enough to encircle the arm)
- If anaeroid, recalibrate periodically before use

Measurements (Plot On Growth Chart)

Weight

Length or Height

Head Circumference

Chest cirumferance

weight	
Age	**Average weight, kg**
• At birth	➢ 3.25
• 3-12 months	➢ $\frac{\text{Age (months)} + 9}{2}$
• 1- 6 years	➢ Age (year) x 2+8
• 7-12 years	➢ $\frac{(\text{Age (year)} \times 7)-5}{2}$

Height/ length	
Age	**Average, cm**
• At birth	➢ 50
• At 1 year	➢ 75
• At 2 year	➢ 87.5
• 3-12 years	➢ (Age (year) x 6) +77 or (Age (year) x 5) + 80

Measuring the lenghth:

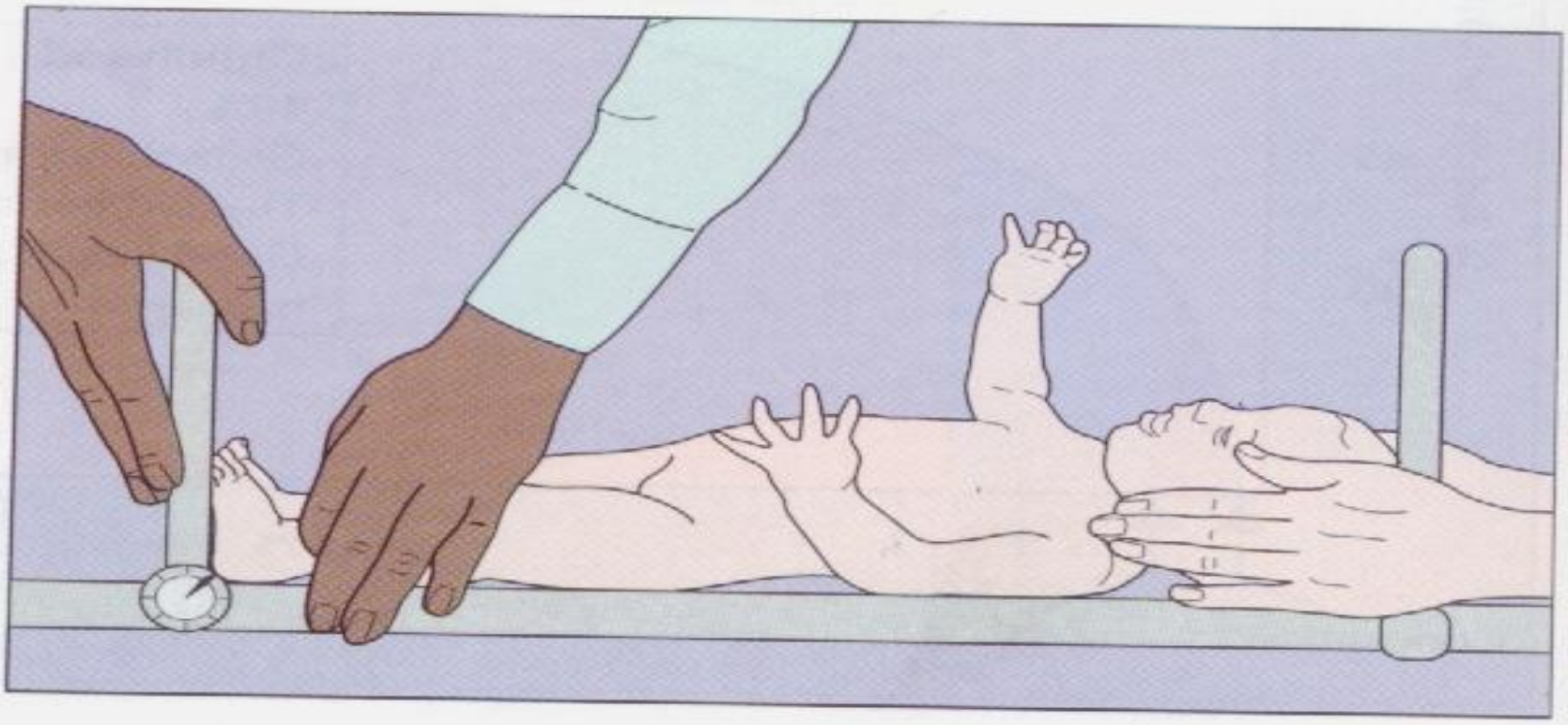
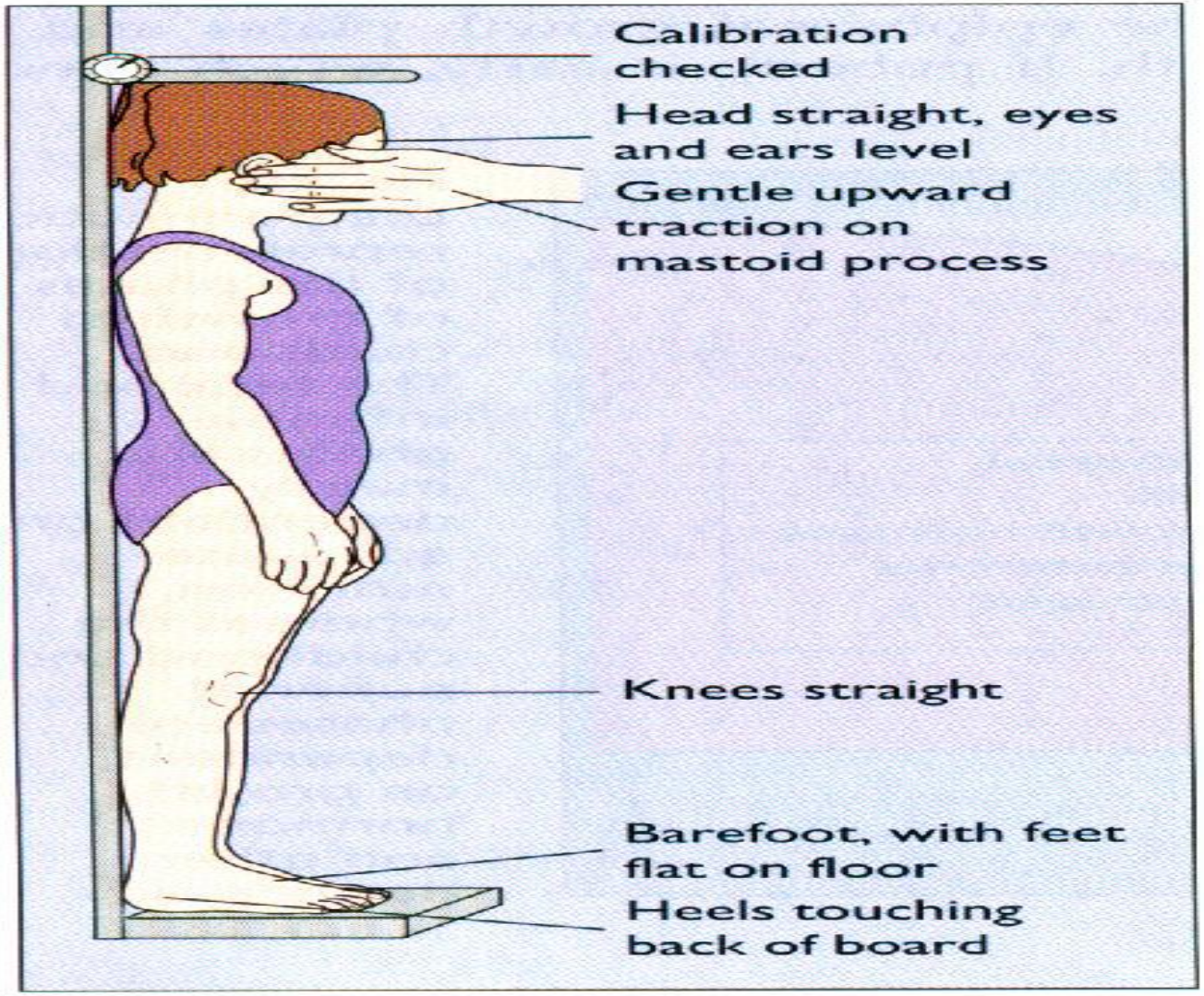


Fig. 10.3 Measuring length in infants and young children. An assistant is required to hold the legs straight.

Measuring the height:



Head circumference (HC)	
Age	Average HC, cm
➢ At birth	➢ 35
➢ At 6 months	➢ 44
➢ At 1 year	➢ 47
➢ At 2 year	➢ 49
➢ 2 - 7 year: increases by ½ cm / year.	
➢ 8 - 12 year: increases by ⅓ cm / year.	

Measuring head circumferance by measuring the greatest circumferance bet. Occipital and frontal prominance.



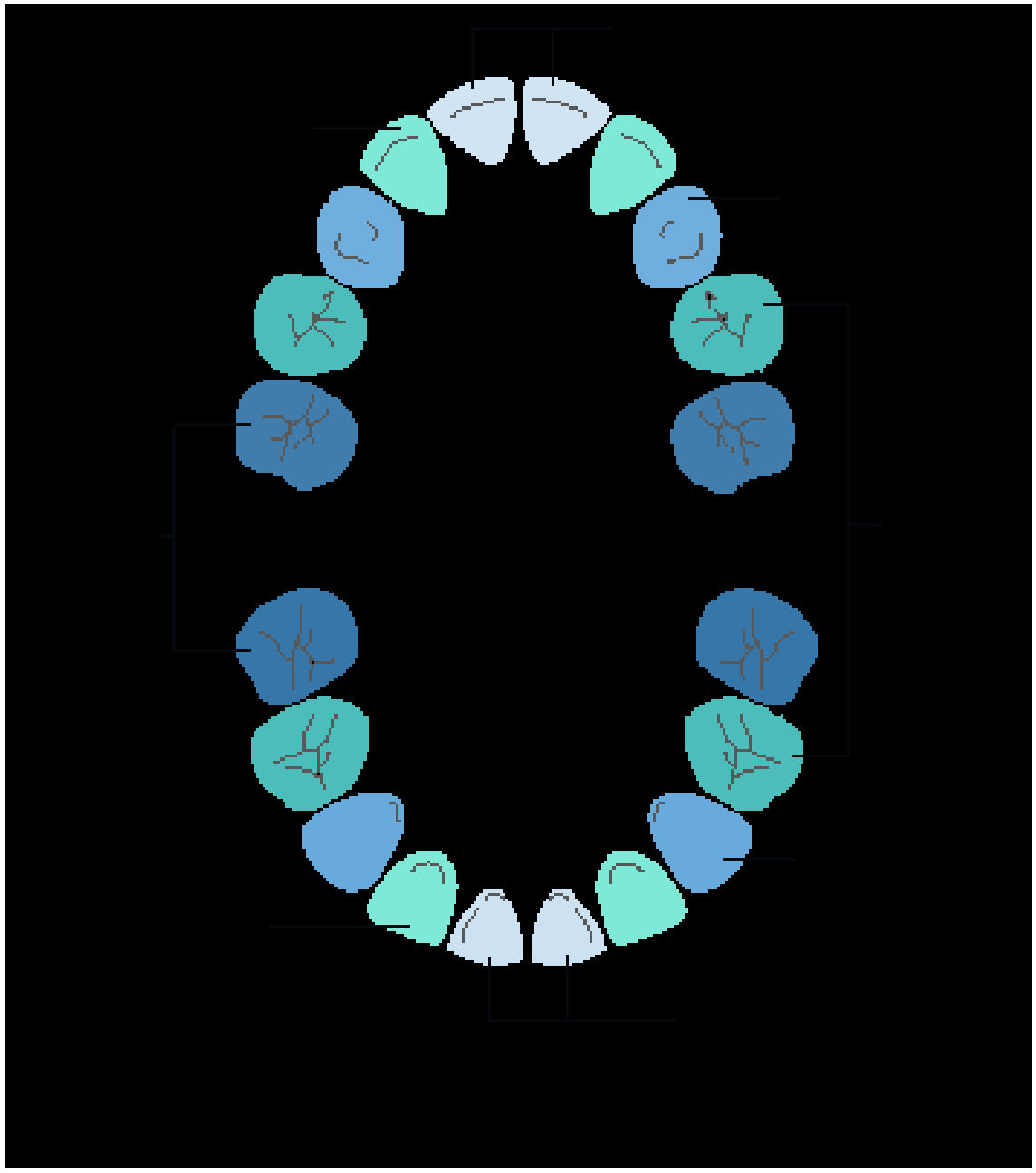
Chest circumference	
Age	Average, cm
At birth	33
1 year	47
• 1-10 years: yearly gain 1.5 cm	
• 11-15 years: yearly gain 3 cm	
• At birth HC > CC	
• 9-12 months HC = CC	
• After 1 year CC > HC	

III -TEETHING

(A)Deciduous (Milk) Teeth :

- They are 20 in number (10 in each Jaw)
- They appear in the following order :

-Central incisors	6 -8 mon
-Lateral incisors	10 –12 mon
-First molar	12 –14 –mon
-Canines	16 –18 mon
-Second molars	20-22 mon



(B) Permanent teeth :

- They are 32 in number (16 in each Jaw)
- They appear in the following order:

-First molar 6 years

-Central incisor 7 years

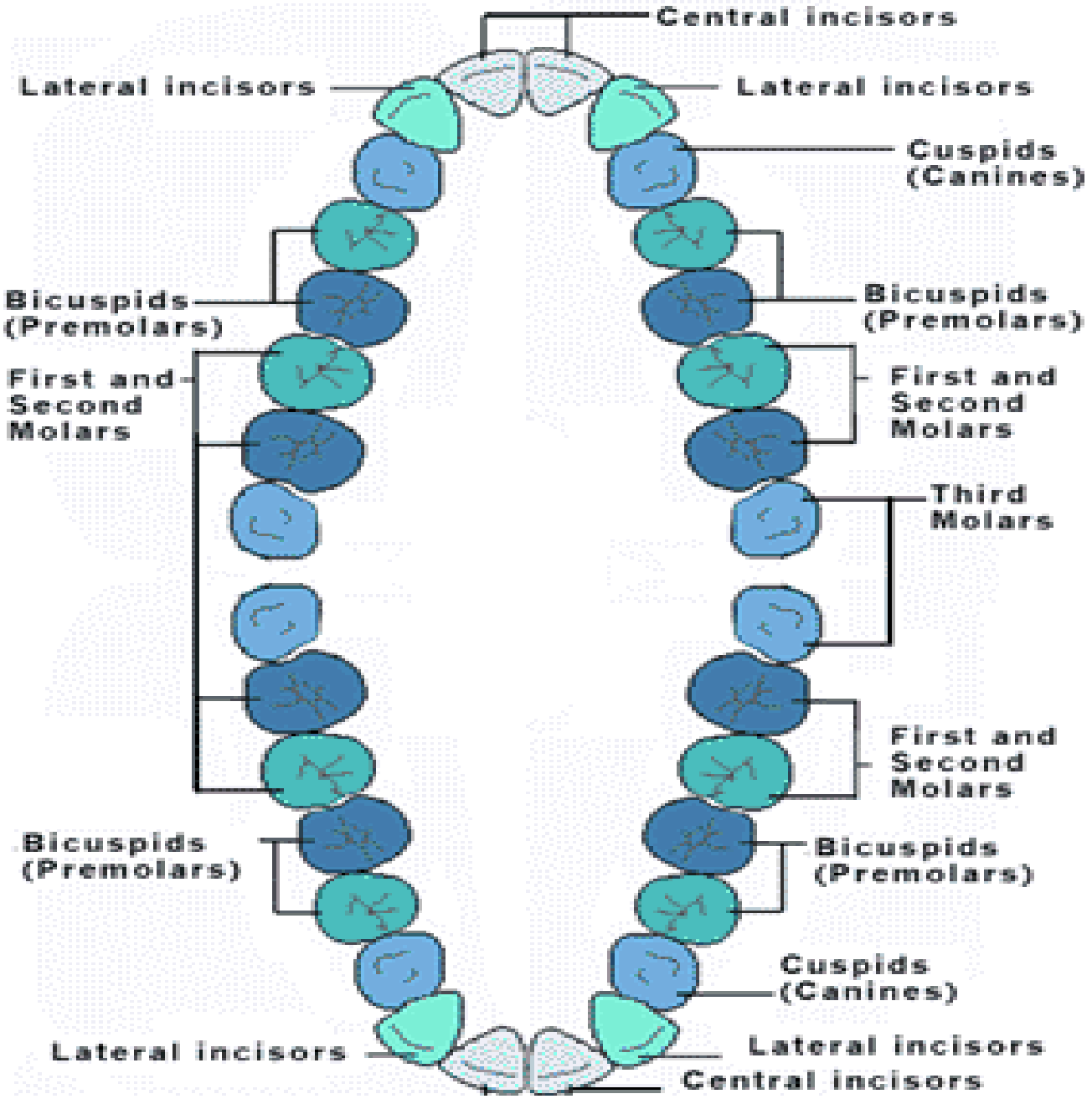
-Lateral incisors 8 years

-First premolar 10 years

-Canine 11 years

-Second molar 12 years

-Wisdom teeth 18-25 years


Central incisors
Lateral incisors
Lateral incisors
Cuspids (Canines)
Bicuspids (Premolars)
Bicuspids (Premolars)
First and Second Molars
First and Second Molars
Third Molars
First and Second Molars
Bicuspids (Premolars)
Bicuspids (Premolars)
Cuspids (Canines)
Lateral incisors
Lateral incisors
Central incisors

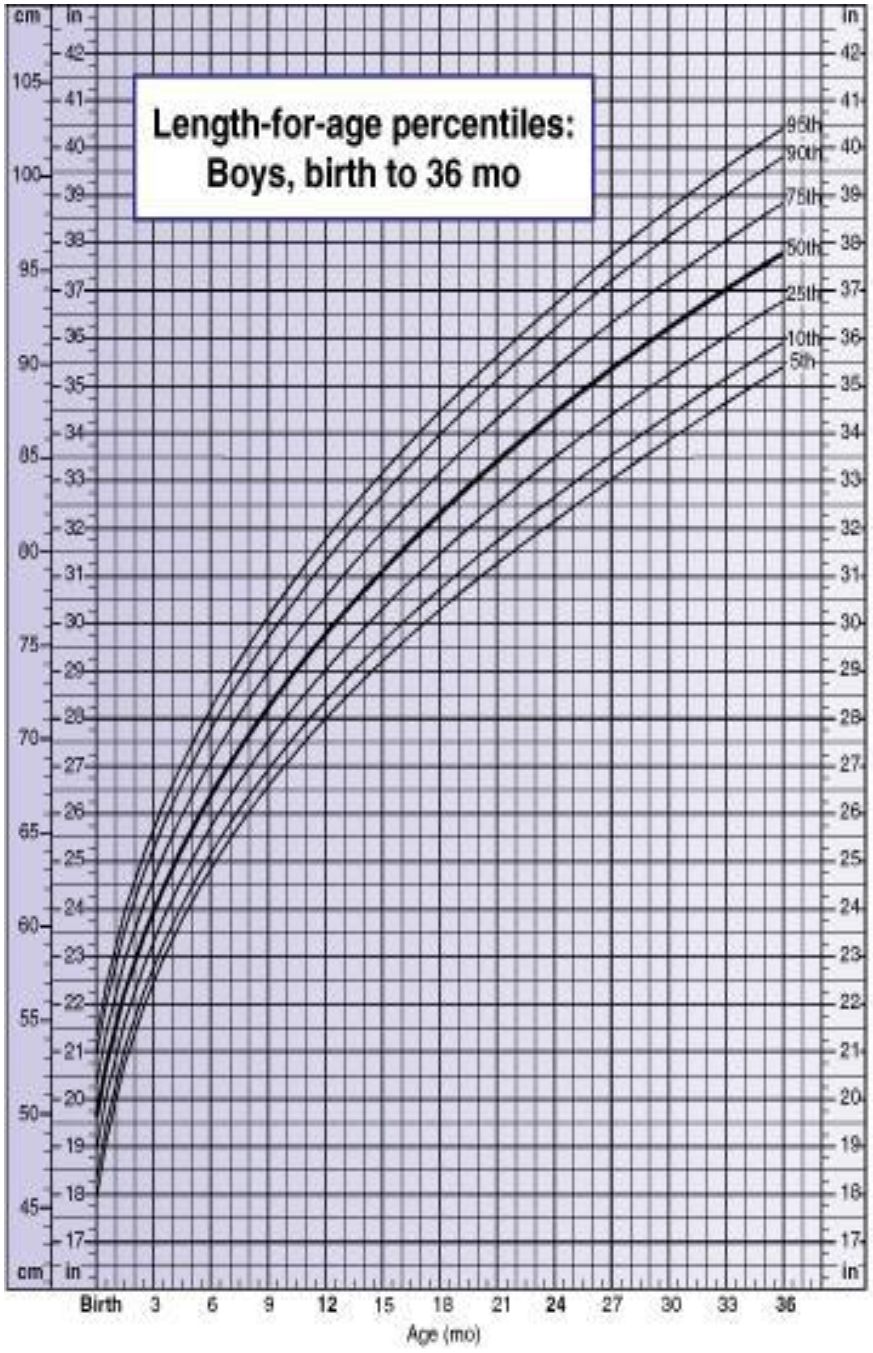
CHILDREN NEED EVALUATION

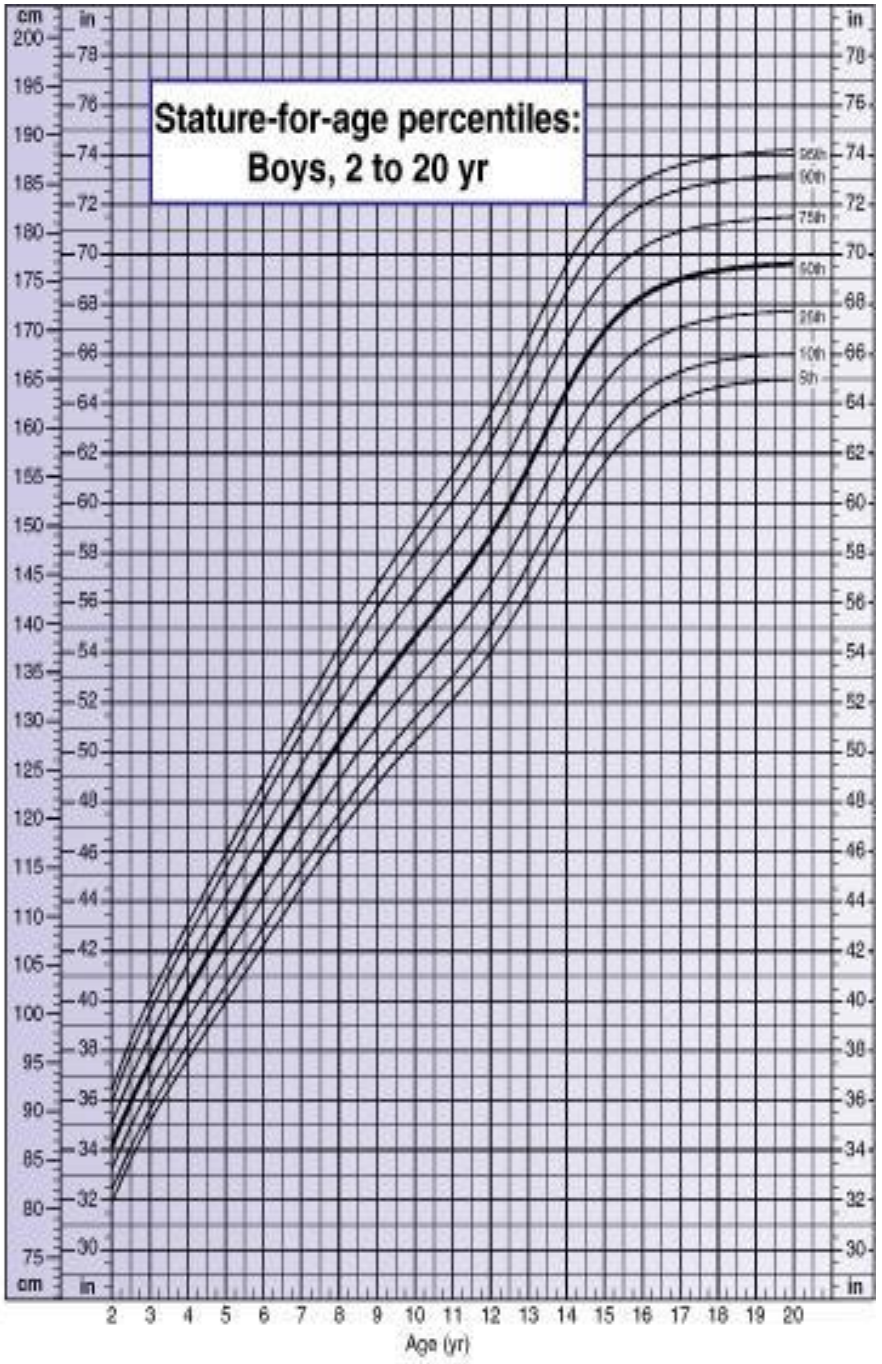
Those with 2 S.D less than or above 95th % or less than 5th %

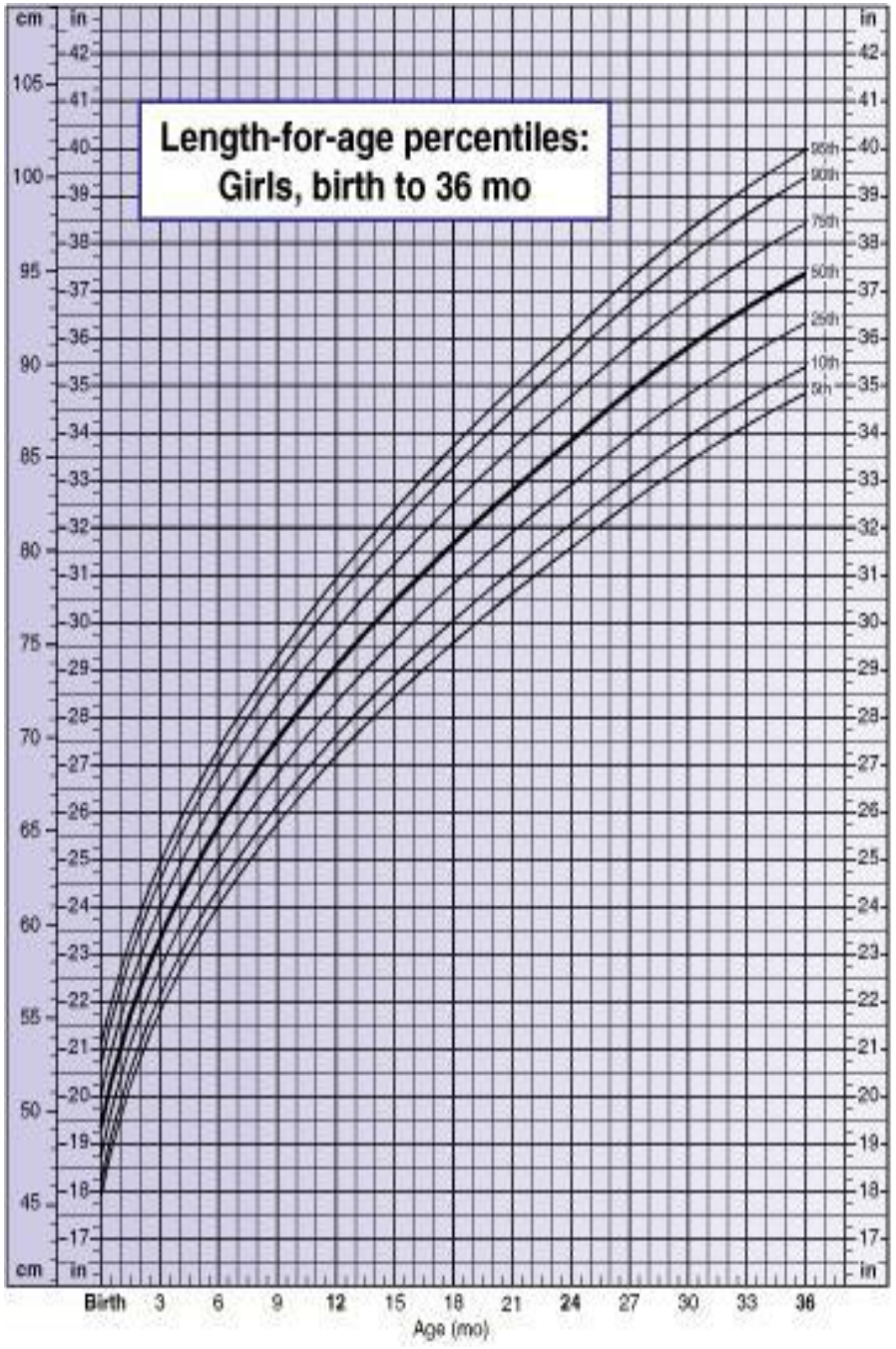
Those with disproportionate growth chart

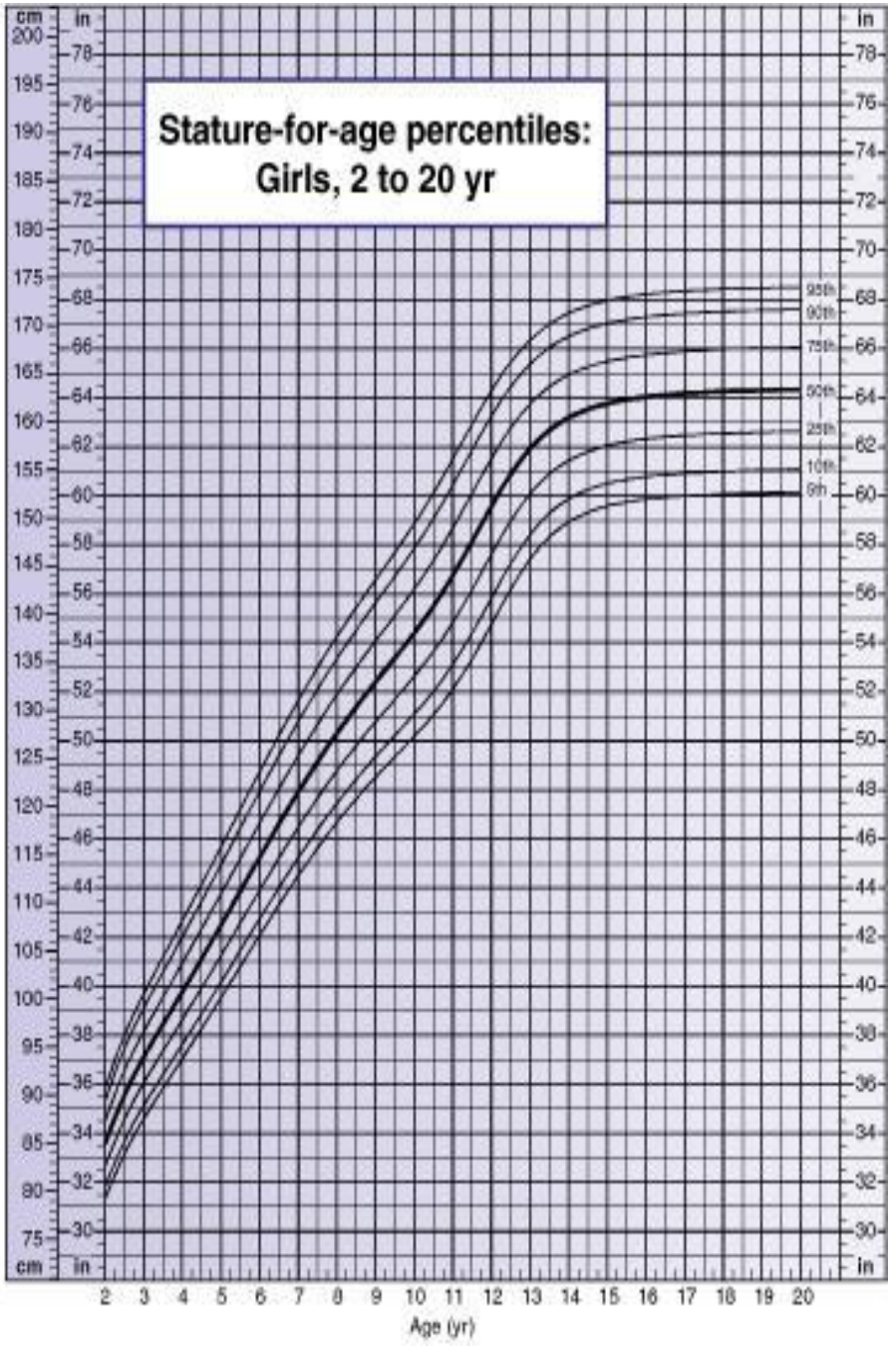
Those with abnormal growth pattern.

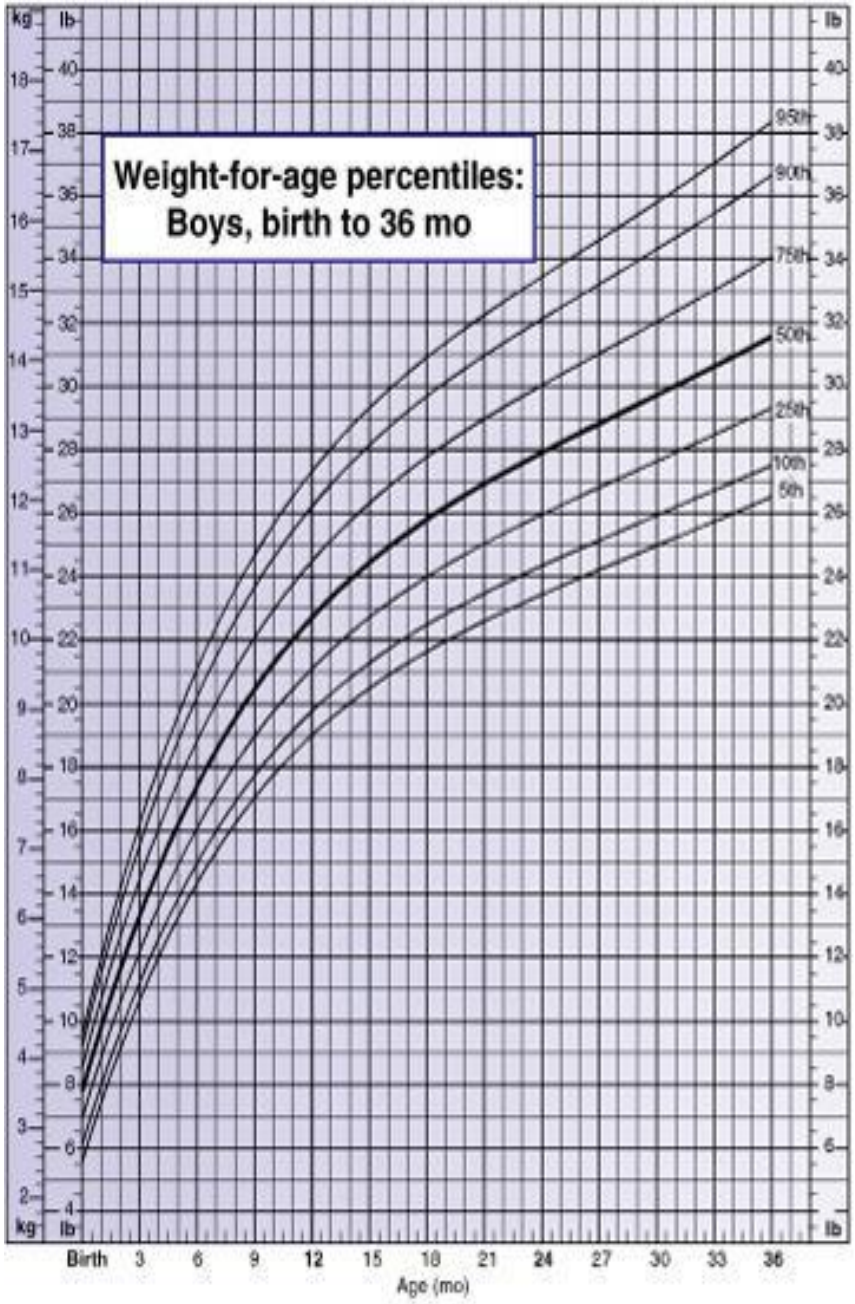

GROWTH CHARTS

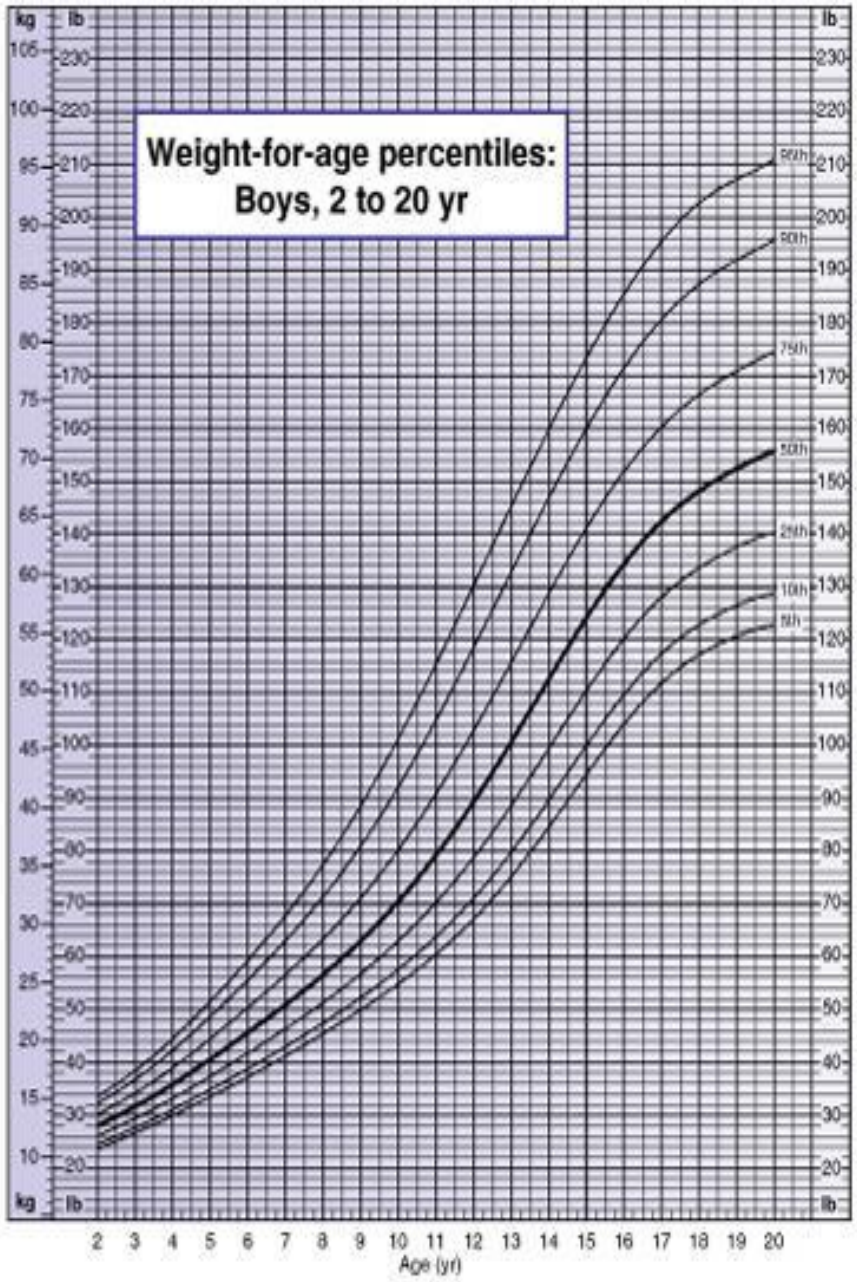

Length-for-age percentiles:
Boys, birth to 36 mo
95th
90th
75th
50th
25th
10th
5th
cm
in
Birth
3
6
9
12
15
18
21
24
27
30
33
36
Age (mo)


Stature-for-age percentiles:
Boys, 2 to 20 yr
95th
90th
75th
50th
25th
10th
5th
cm
in
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
Age (yr)

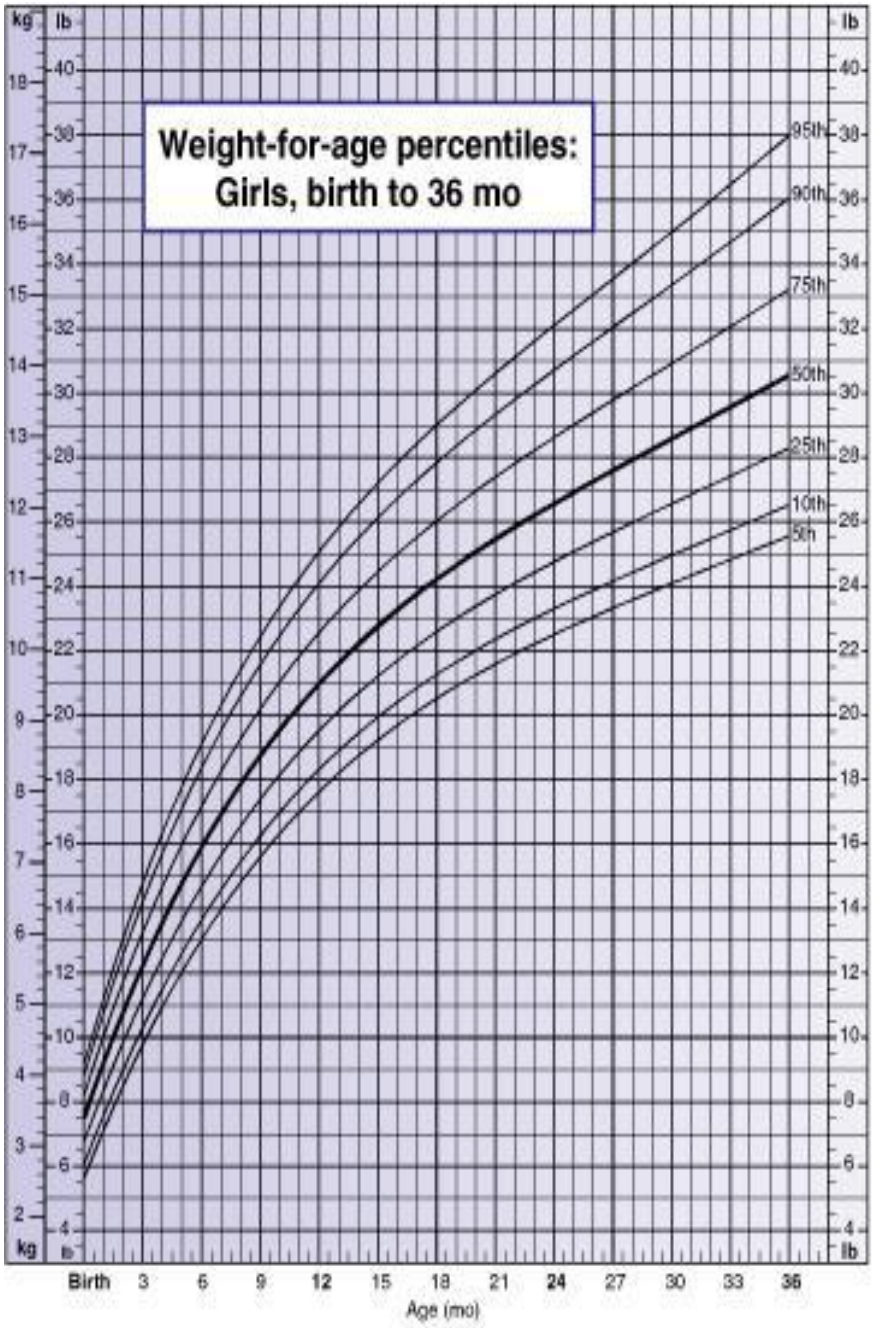

Length-for-age percentiles:
Girls, birth to 36 mo
cm
in
95th
90th
75th
50th
25th
10th
5th
Birth 3 6 9 12 15 18 21 24 27 30 33 36
Age (mo)

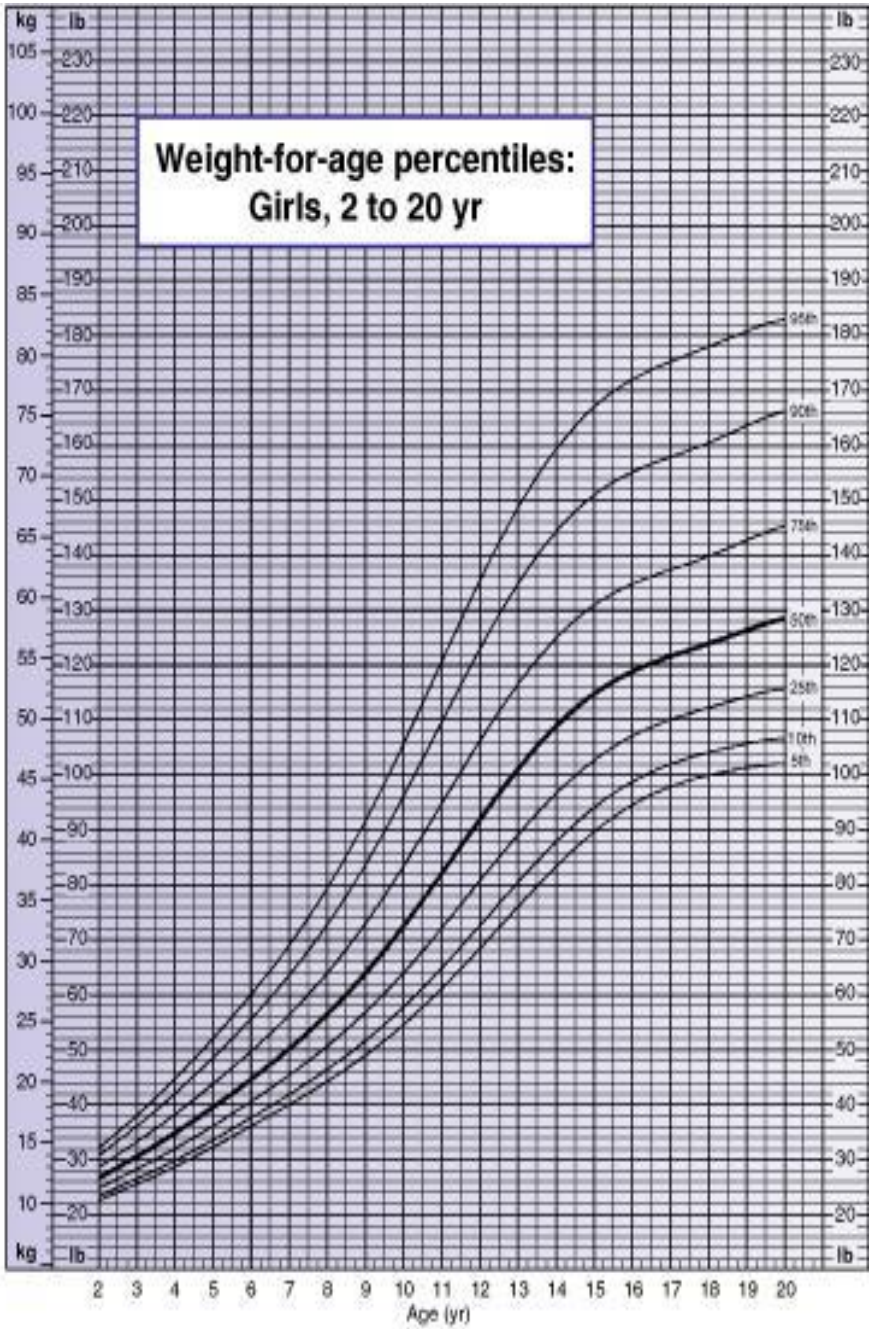

Stature-for-age percentiles:
Girls, 2 to 20 yr
cm
in
95th
90th
75th
50th
25th
10th
5th
2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
Age (yr)


Weight-for-age percentiles:
Boys, birth to 36 mo
kg
lb
95th
90th
75th
50th
25th
10th
5th
Birth
3
6
9
12
15
18
21
24
27
30
33
36
Age (mo)


Weight-for-age percentiles:
Boys, 2 to 20 yr
kg
lb
95th
90th
75th
50th
25th
10th
5th
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
Age (yr)

A

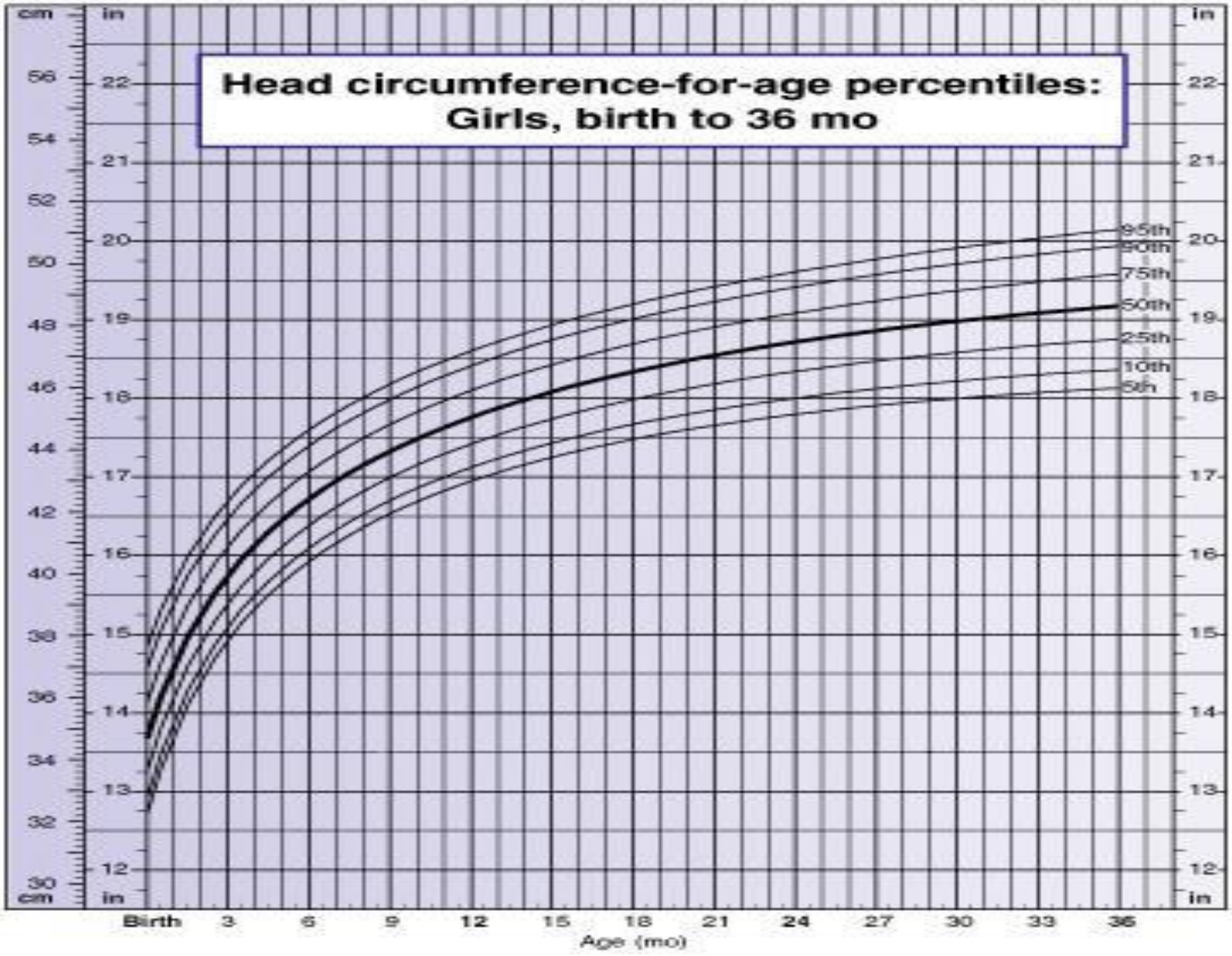

Weight-for-age percentiles:
Girls, birth to 36 mo
95th
90th
75th
50th
25th
10th
5th
kg
lb
Birth
Age (mo)


Weight-for-age percentiles:
Girls, 2 to 20 yr
95th
90th
75th
50th
25th
10th
5th
kg
lb
Age (yr)

B


Head circumference-for-age percentiles:
Boys, birth to 36 mo
cm
in
56
54
52
50
48
46
44
42
40
38
36
34
32
30
22
21
20
19
18
17
16
15
14
13
12
95th
90th
75th
50th
25th
10th
5th
Birth
3
6
9
12
15
18
21
24
27
30
33
36
Age (mo)

A


Head circumference-for-age percentiles:
Girls, birth to 36 mo
cm
in
56
54
52
50
48
46
44
42
40
38
36
34
32
30
22
21
20
19
18
17
16
15
14
13
12
95th
90th
75th
50th
25th
10th
5th
Birth
3
6
9
12
15
18
21
24
27
30
33
36
Age (mo)

Head examination:

*pallor, jaundice, cyanosis

*normal or abnormal shape :box shape in rickets

*Fontanel size if open : wide in rickets

*hair: silky..., coarse brittle...., easily detachable, flag sign....

* Normal or abnormal facies.

Mongolian facies…down syndrome

thalassemic facies ... Thalassemia

*Eye, nose, ear ex: puffy or sunken eyes and if there is any abnormality like proptosis, depressed nasal bridge, low set ears

*skin & Mucous membranes :

-pallor: anemia

-cyanosis: respiratory failure, cyanotic heart dis

- rash

Central cyanosis:



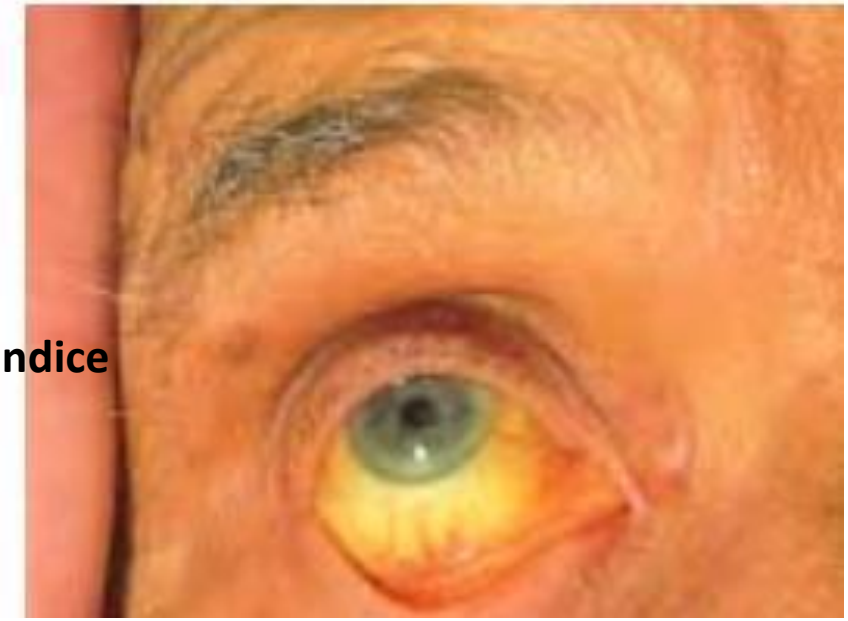
Central cyanosis:



jaundice



Normal vs jaundice

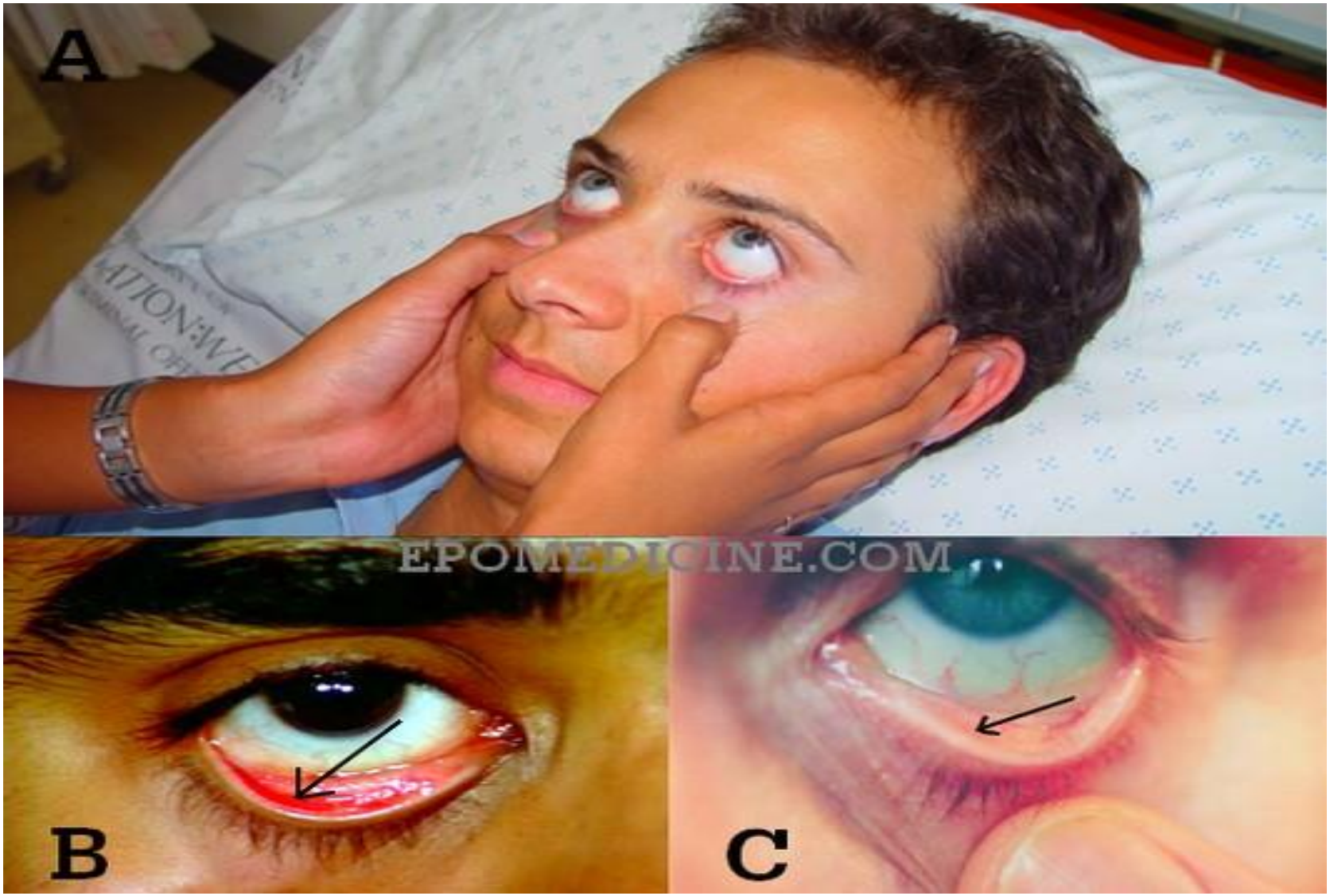


jaundice





pallor



pallor



pallor



FONTANELLES :

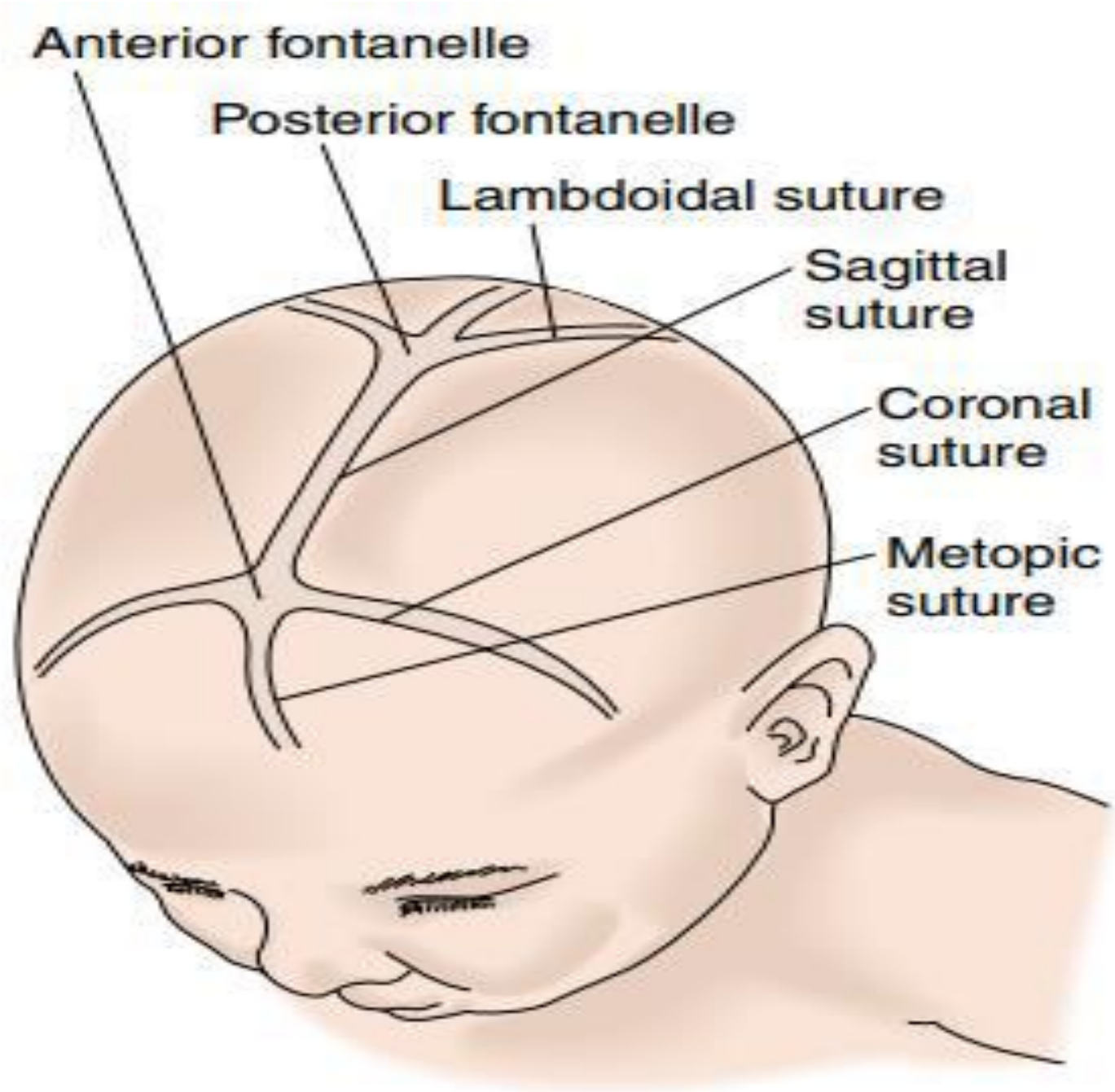
- There are 6 fontanelles : (1. Anterior & 1. Posterior 2. Sphenoid and 2 mastoid)
- The anterior & Posterior ones are clinically important.

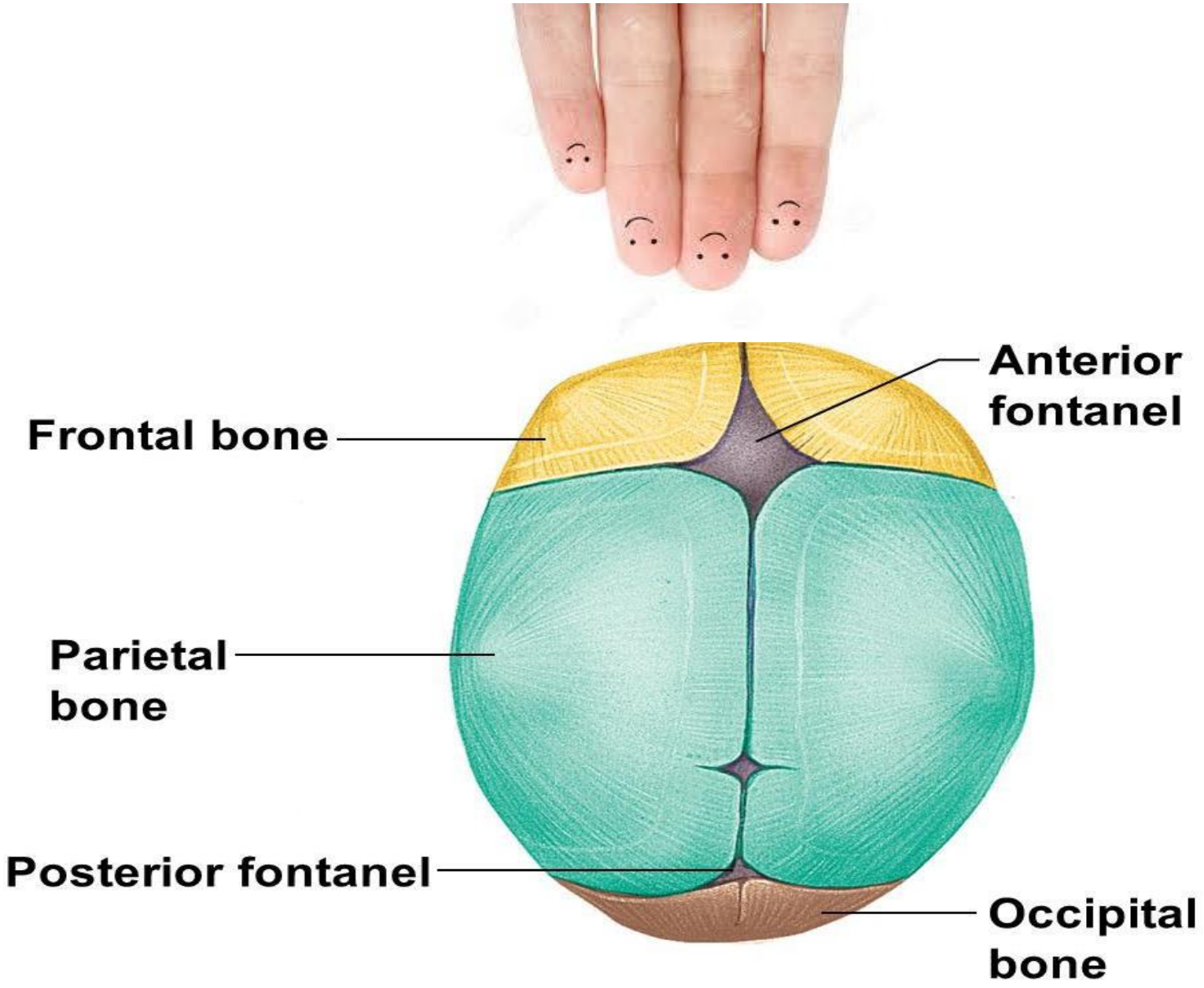
*The post. Fontanelle : closes at full term or 1-2 mon later

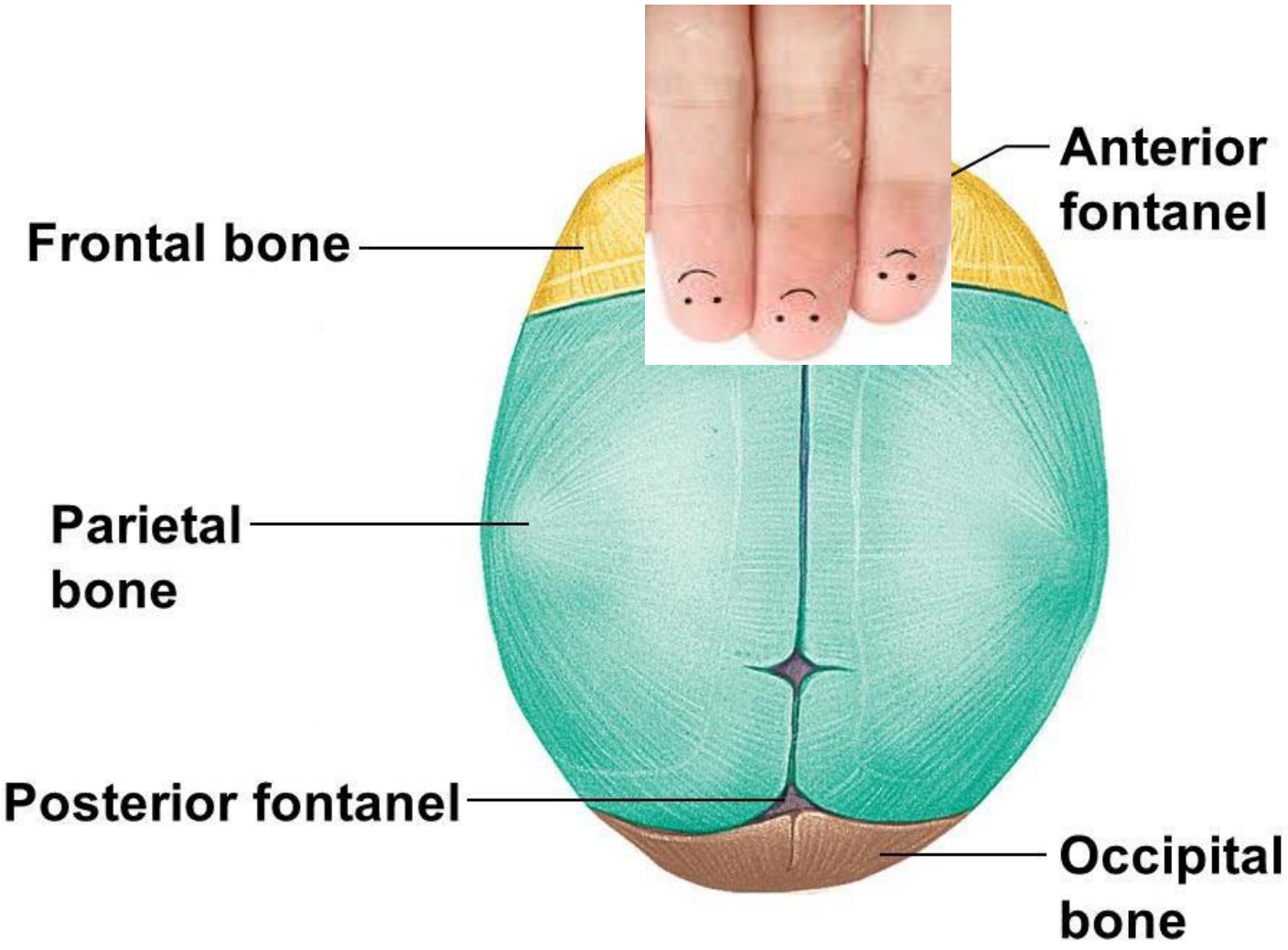
*The Ant. Fontenelle : is 3 fingers at birth (average 4-5cm)

*2 fingers at 6 month

*1 finger at 1 year. closed at 1.5yr


Anterior fontanelle
Posterior fontanelle
Lambdoidal suture
Sagittal suture
Coronal suture
Metopic suture


Anterior fontanel
Frontal bone
Parietal bone
Posterior fontanel
Occipital bone


Anterior fontanel
Frontal bone
Parietal bone
Posterior fontanel
Occipital bone

Puffy eyes





Puffy eyes



Hydrocephlus:



*Increased head circumferanc

*Separation of sutures

*Tense large anterior fontanel

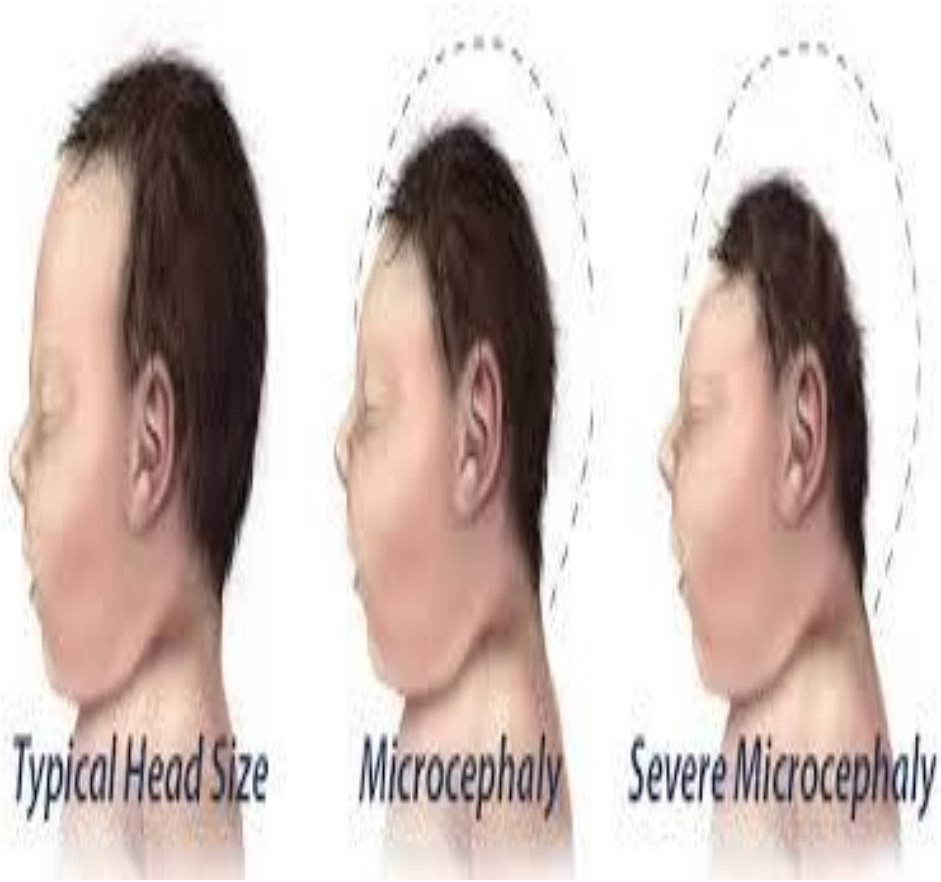
*Persistent posterior fontanel

*Widely opened fontanels

*Scalp veins may be dilated

*Sun set eye apperance

Microcephaly:



*Decrease craniofacial dispropotion.

*May closed sutures (Craniocynstosis)

*May appear signs of increased intracranial tension in craniocynstosis.)

Proptosis:



Thalassemic facies:



*Frontal and parietal bossing

*Prominent maxilla

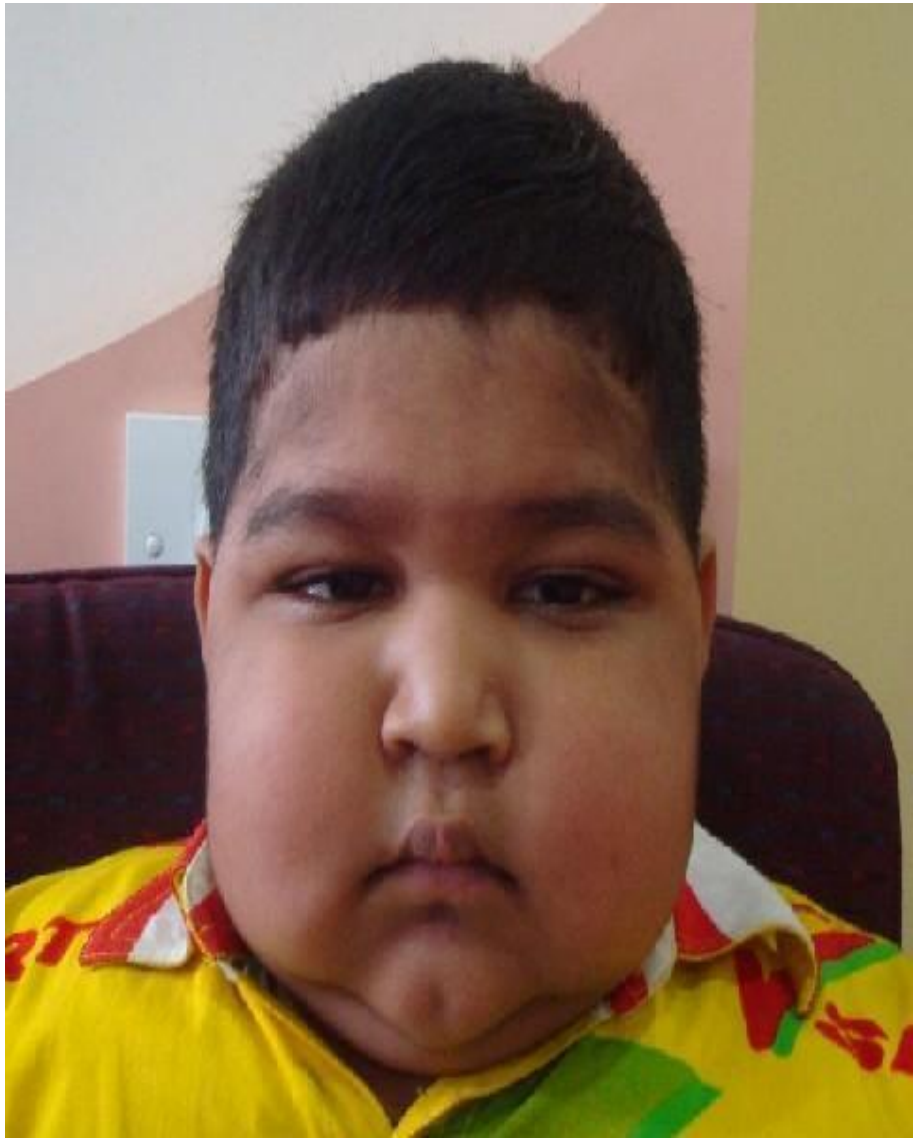
*Exposure of upper central incisors and apparently depressed nasal bridge.

*pallor, jaundice, hemosiderosis combined to give greenish brown skin.

Cushinoid facies:



Cushinoid face:



Causes:

*cushing syndrome

*iatrogenic corticosteroid

Down facies (mongolian facies)

*Upward and outward slanting of eye.

*Medial epicanthic folds.

*Depressed nasal bridge.

*Small mouth cavity with protruded tongue.

*Low set ears.

*Short neck.





Kwashirkor:

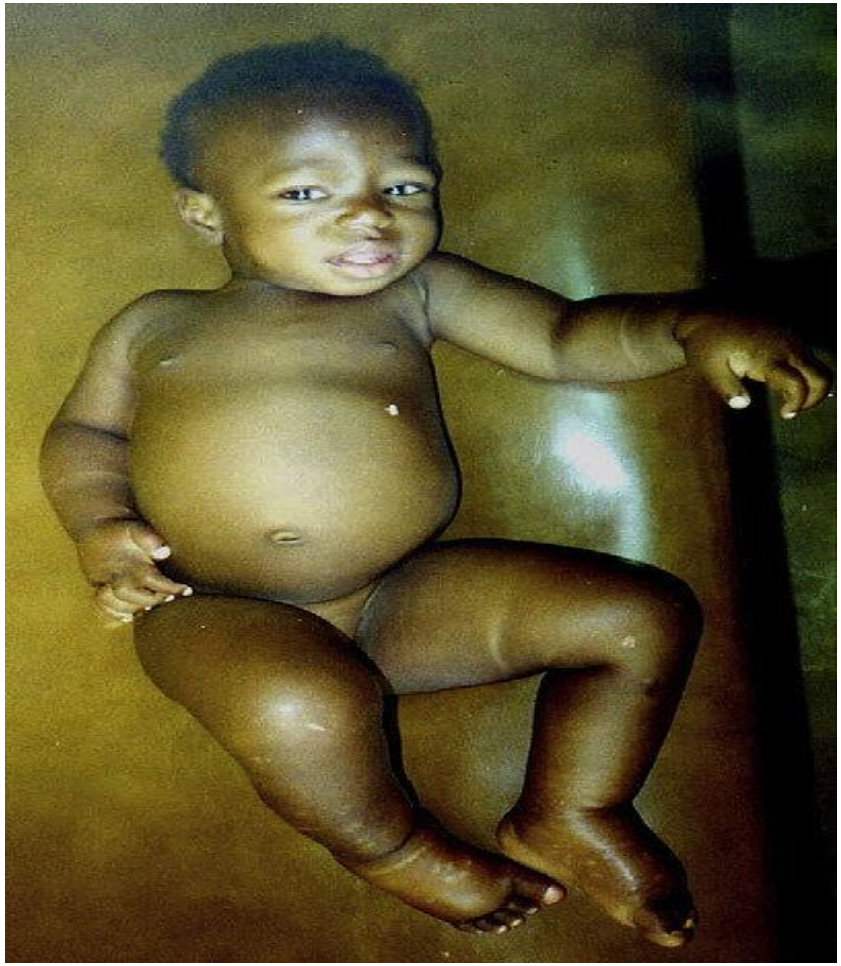
*Apathy

*doll face

*Edema

*Brittle thinning hair with Flag sign

*Carcky paint dermatosis



Marasmus:

Loss of subcutanous fat, muscle from abdomen , trunk, limbs , face (**senile facie**)



Cretinism (congenital hypothyrodisim)

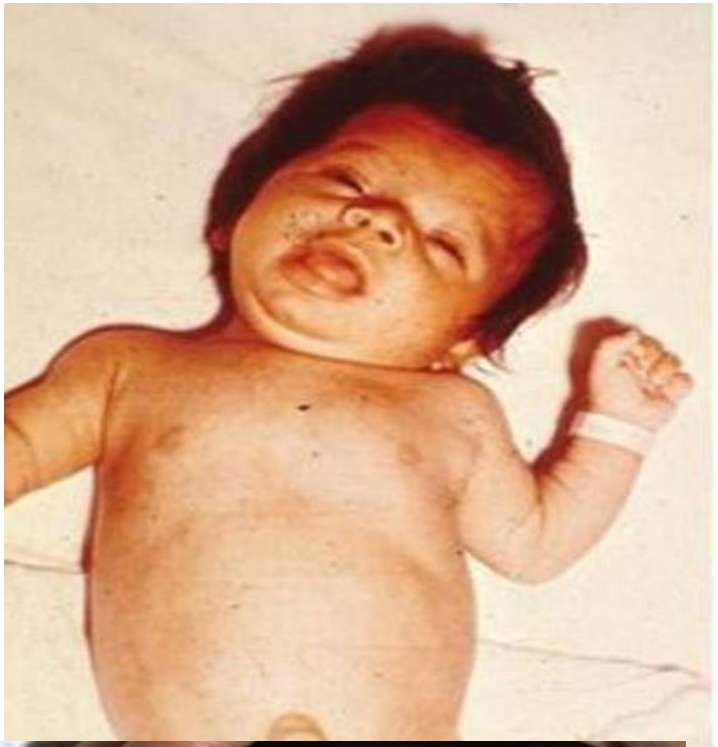
*Coarse brittle hair , hair line reach far down on forehead

*eyes appear faraway with narrow palpebral fissure.

*depressed nasal bridge.

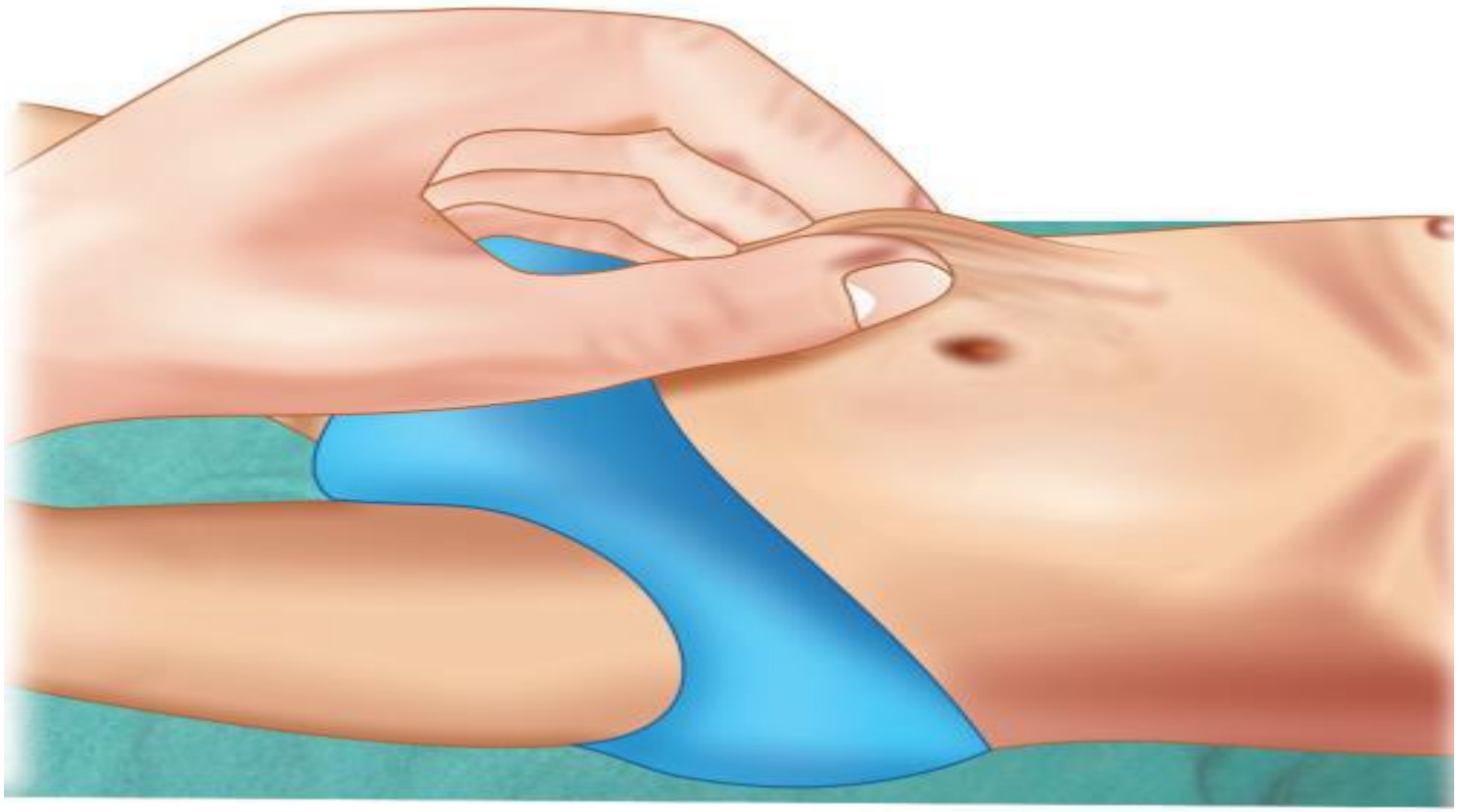
*open mouth with large protruded tongue

*thick , short neck





Method of skin turgor examination:



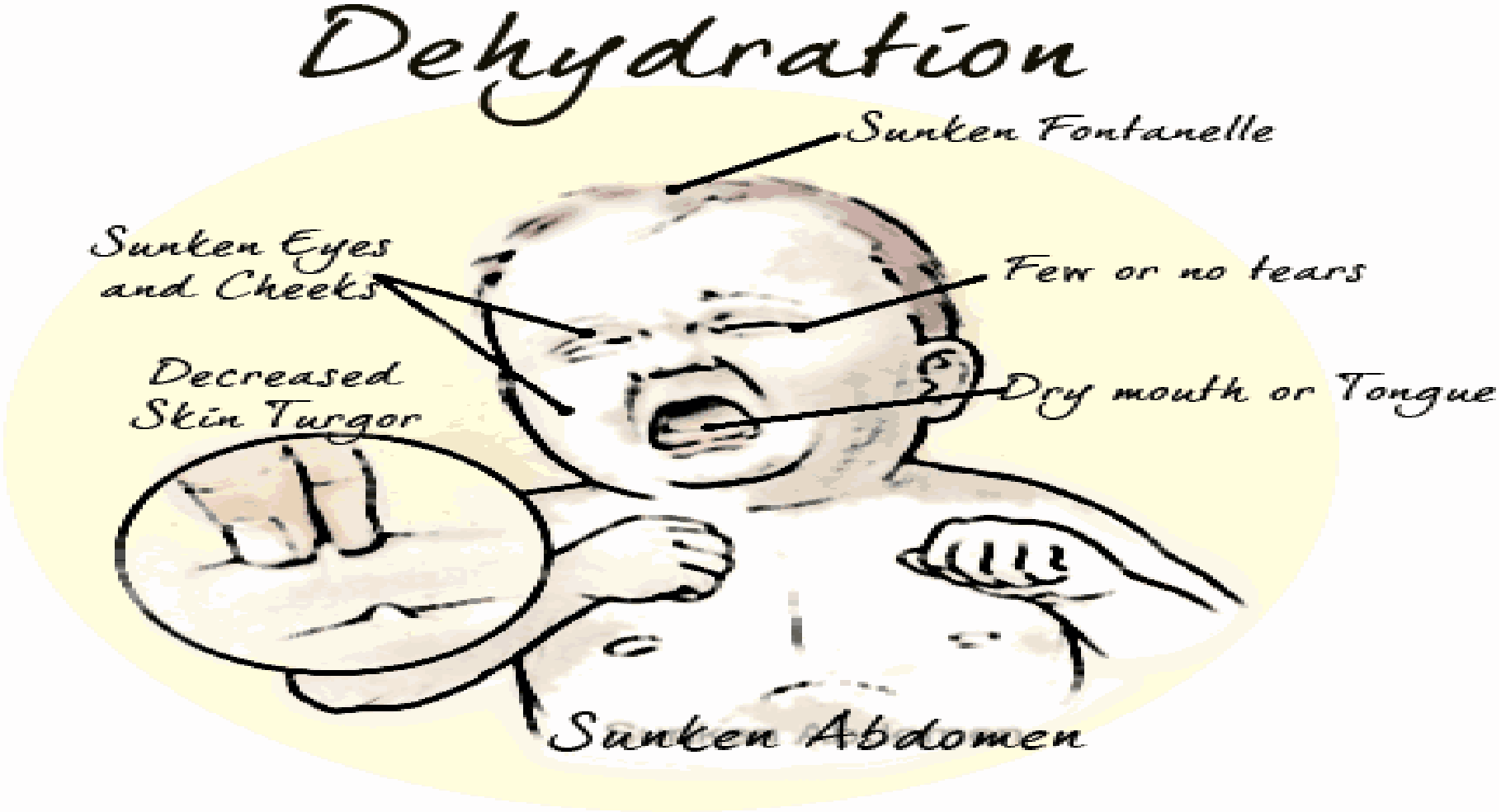
Delayed skin turgor in dehydration:



Dehydrated with sunken eyes:



Signs of dehydration



Oral monaliasis:



Follicular tonsilitis:



Neck examination:

Abnormal position

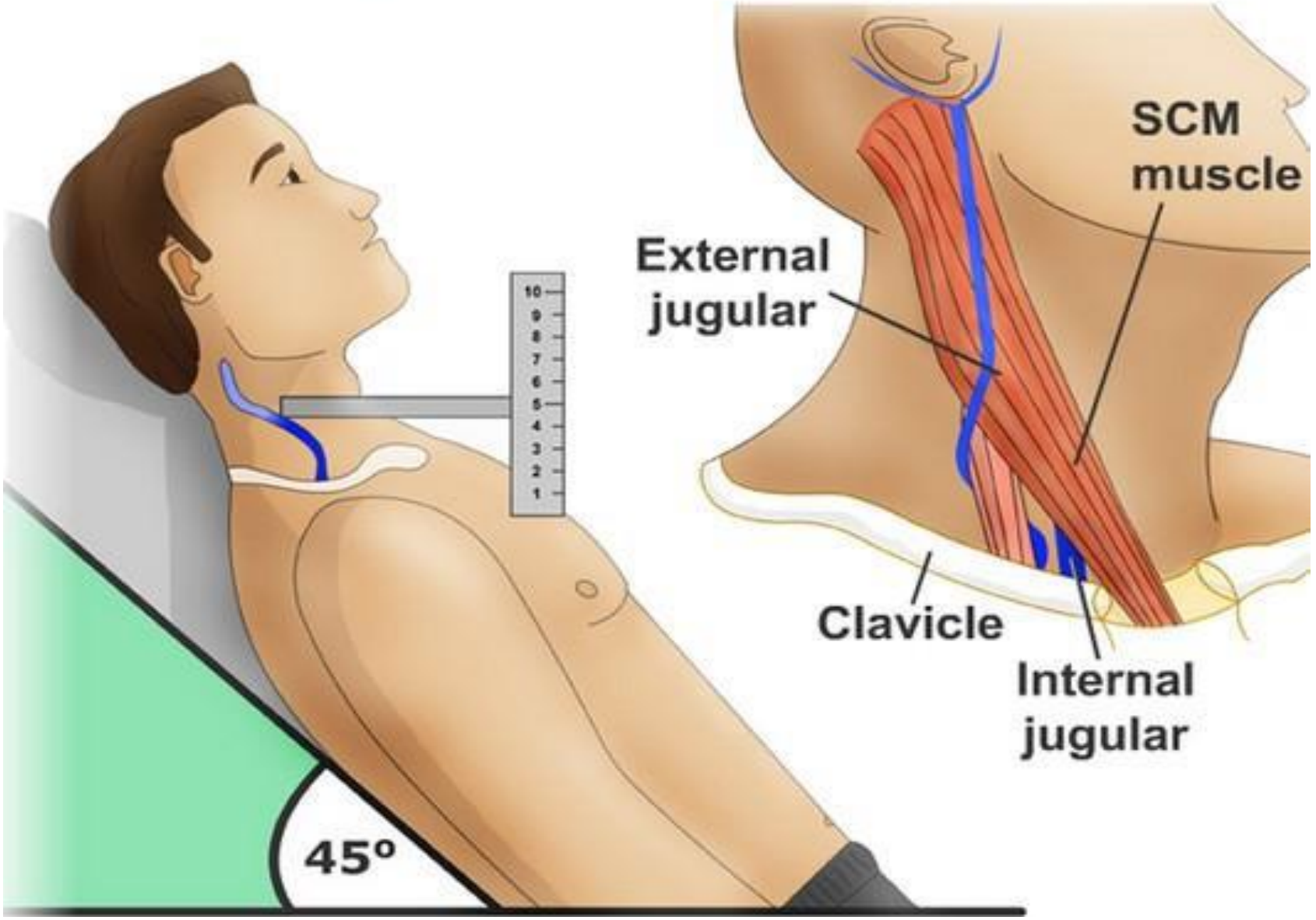
Thyroid gland

Carotid pulse

Lymph nodes enlargement

JVP : congested or not, pulsating or not

Jugular venous pressure



Lymph nodes:

*If abnormal is size or texture record location, consistency, tenderness, size in centimeters.

*Significant lymph node enlargement :

more than 1,5 CM in inguinal region

more than 1 CM in axillary

more than 2 CM ant. Cervical

more than .5 CM in supraclavicular and epitrochlear

*Tender if inflammation (acute)

*not tender if malignant

*Matted : TB

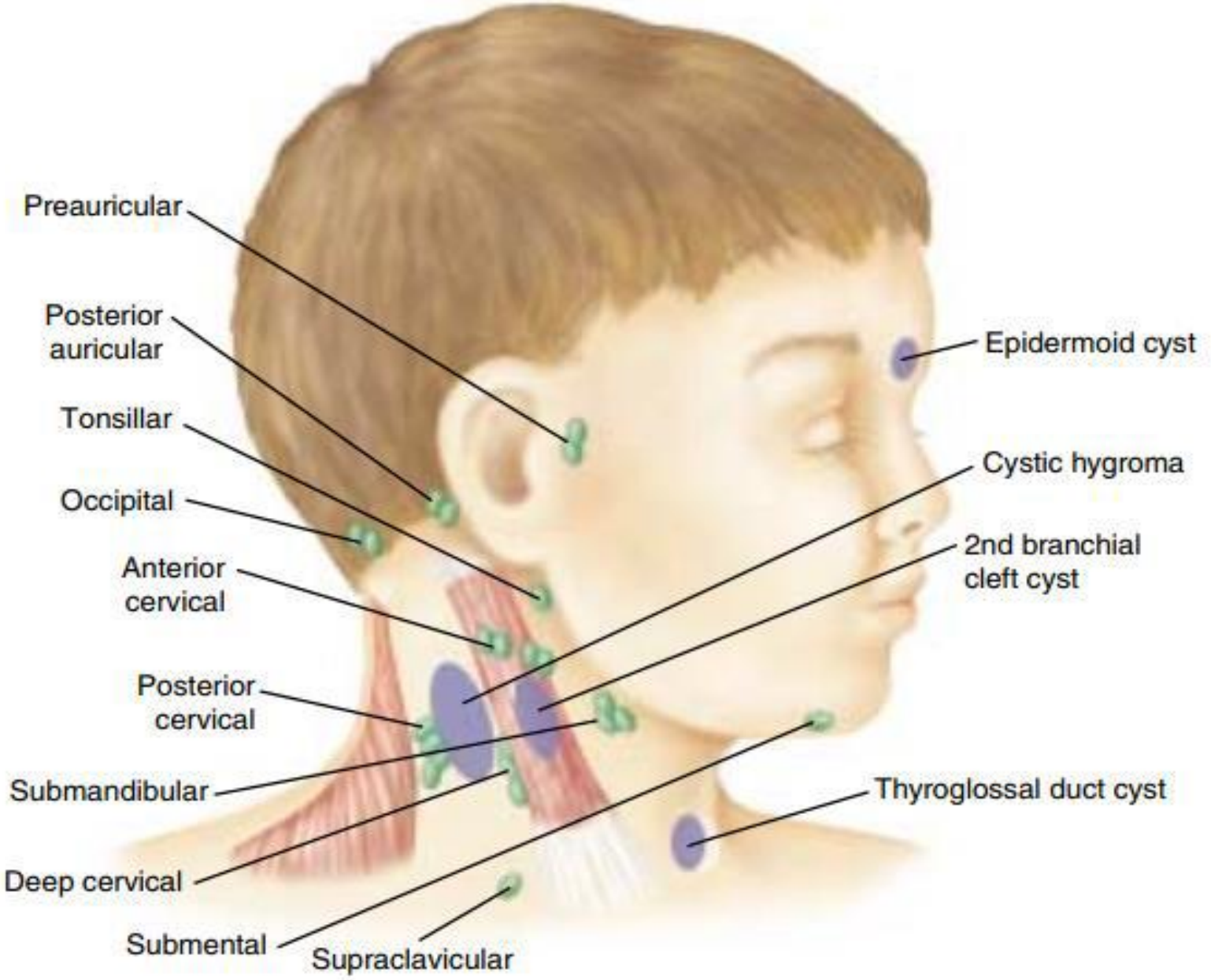
*Fixed: malignancy

*Generalized : TB , leukemia and lymphoma.

*firm or hard : malignancy.

*Regional : infection

*Generalized : more than 2 groups in systemic disease


Preauricular
Posterior auricular
Tonsillar
Occipital
Anterior cervical
Posterior cervical
Submandibular
Deep cervical
Submental
Supraclavicular
Epidermoid cyst
Cystic hygroma
2nd branchial cleft cyst
Thyroglossal duct cyst

Groups of lymph nodes:

- inguinal
- axillary
- epi trochlear
- cervical LNs: circular or vertical.

circular : submental ,submandibular , preauricular, postauricular , occipital.

Vertical:

ant.cervical , post cervical , supraclavicular lymph nodes .

note : Lt supraclavicular LNs (virchows LNs) …are mirror for the intraabdominal problems

Rt supraclavicular LNs: in intrathoracic problems

Upper limbs examination:

Capillary refill
Nails: splinter he (in infective endocarditis),
Spooning
Pallor: in anemia
Clubbing: pale or cyanotic
-pale: infective endocarditis
-cyanotic: congenital cyanotic heartdisease,

Grades of clubbing:

1st: oblitrated angel

2nd: parrot peak

3rd: drumstick

4th: hypertrophic osteoarthropathy

Short fingers(brachydactly), any deformities or swelling, Thumb anomaly
Semian crease, curved little finger(clinodactyly).. (down $)
Osler nodules, Janeway spots (infective endocarditis)
Broadening of distal ends of radius & ulna (rickets)
Skin rash

Peripheral perfusion:

cold cyanotic in poor peripheral perfusion

capillary refill test :

normal: 1 to 2 seconds.

Prolonged if more than 2 sec.

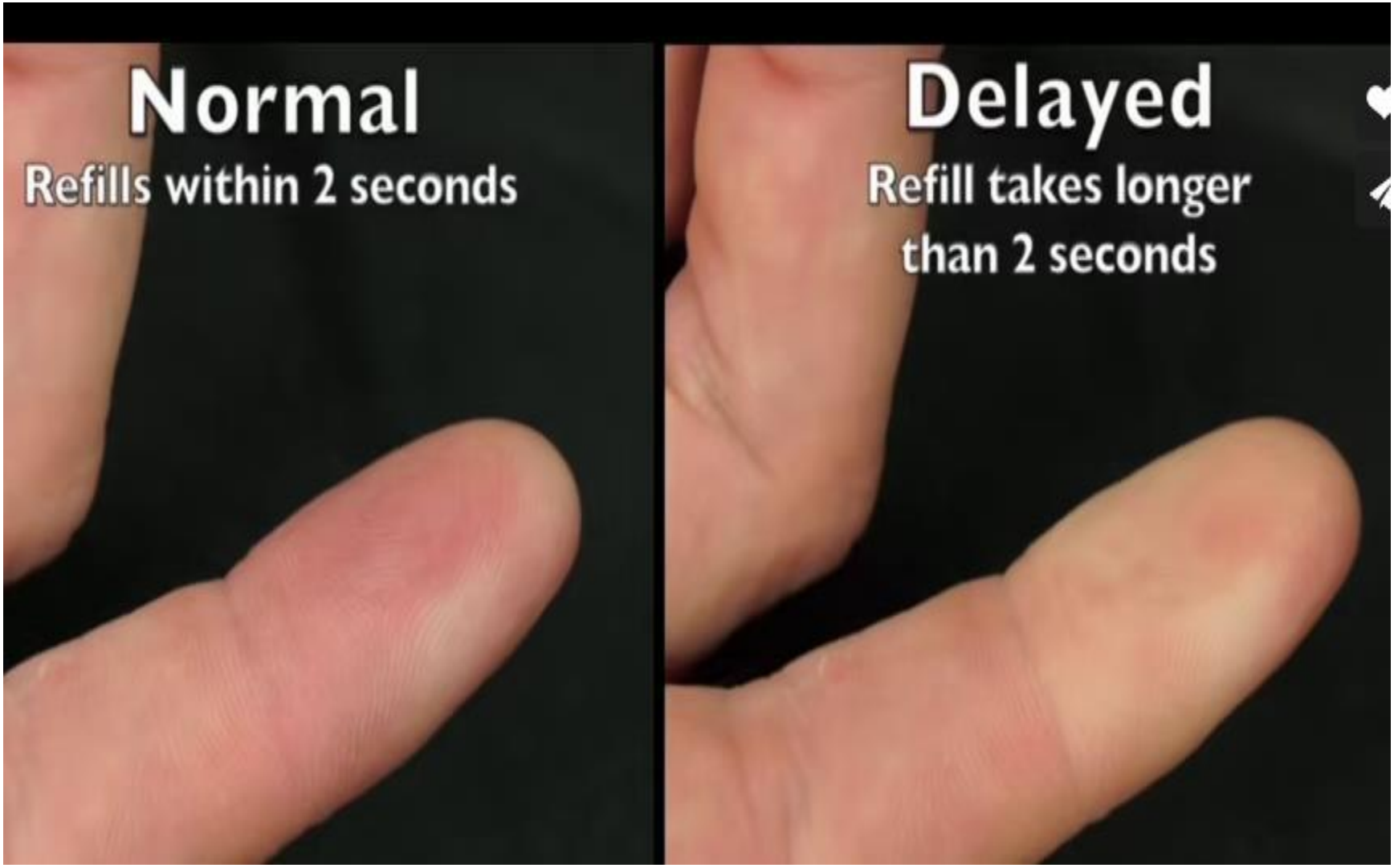
Palpation of the radial pulse:

That felt over the distal part of radial artery at the wrist

Brachial pulse:

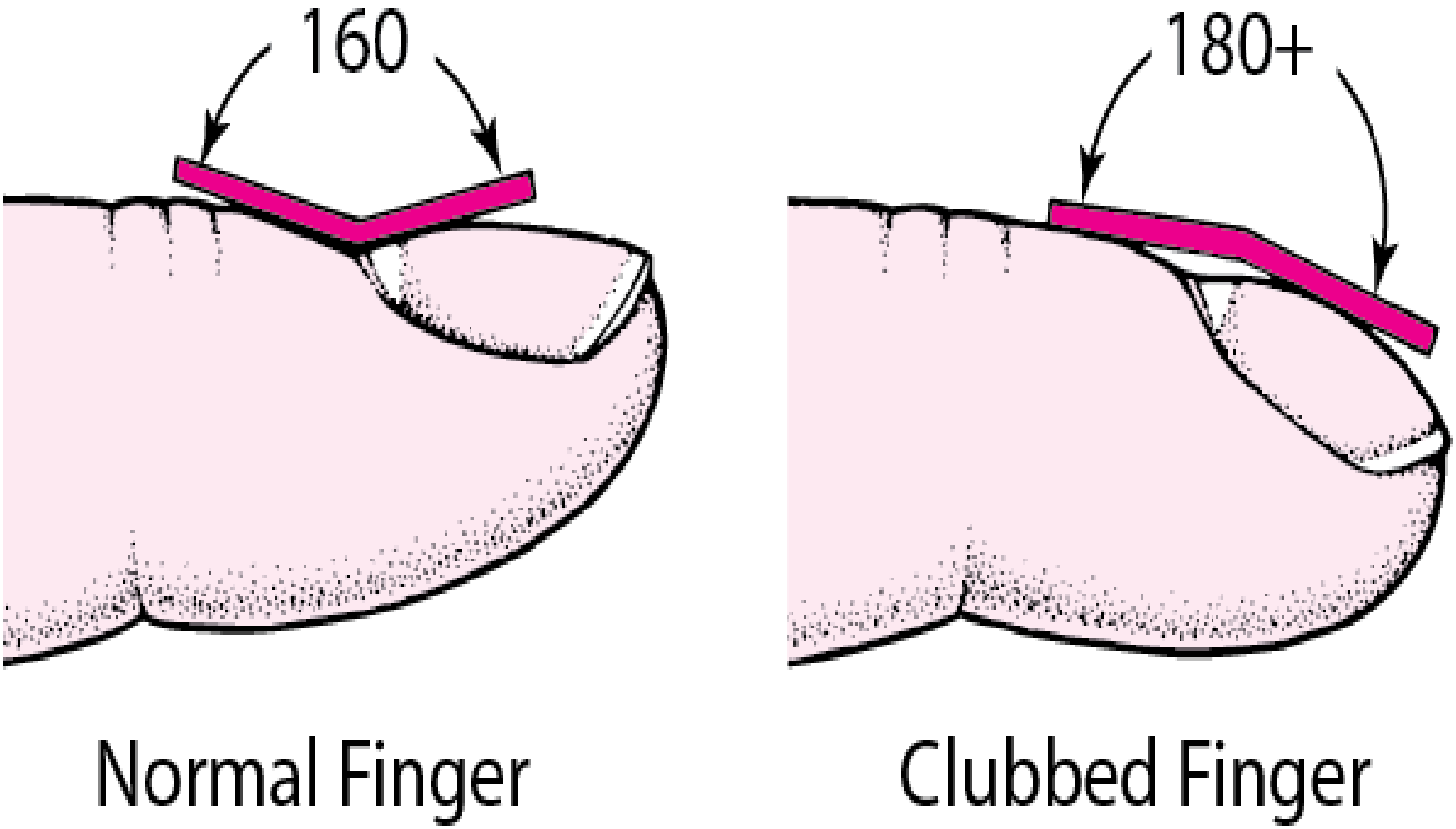
that which is felt over the brachial artery at the inner aspect of the elbow



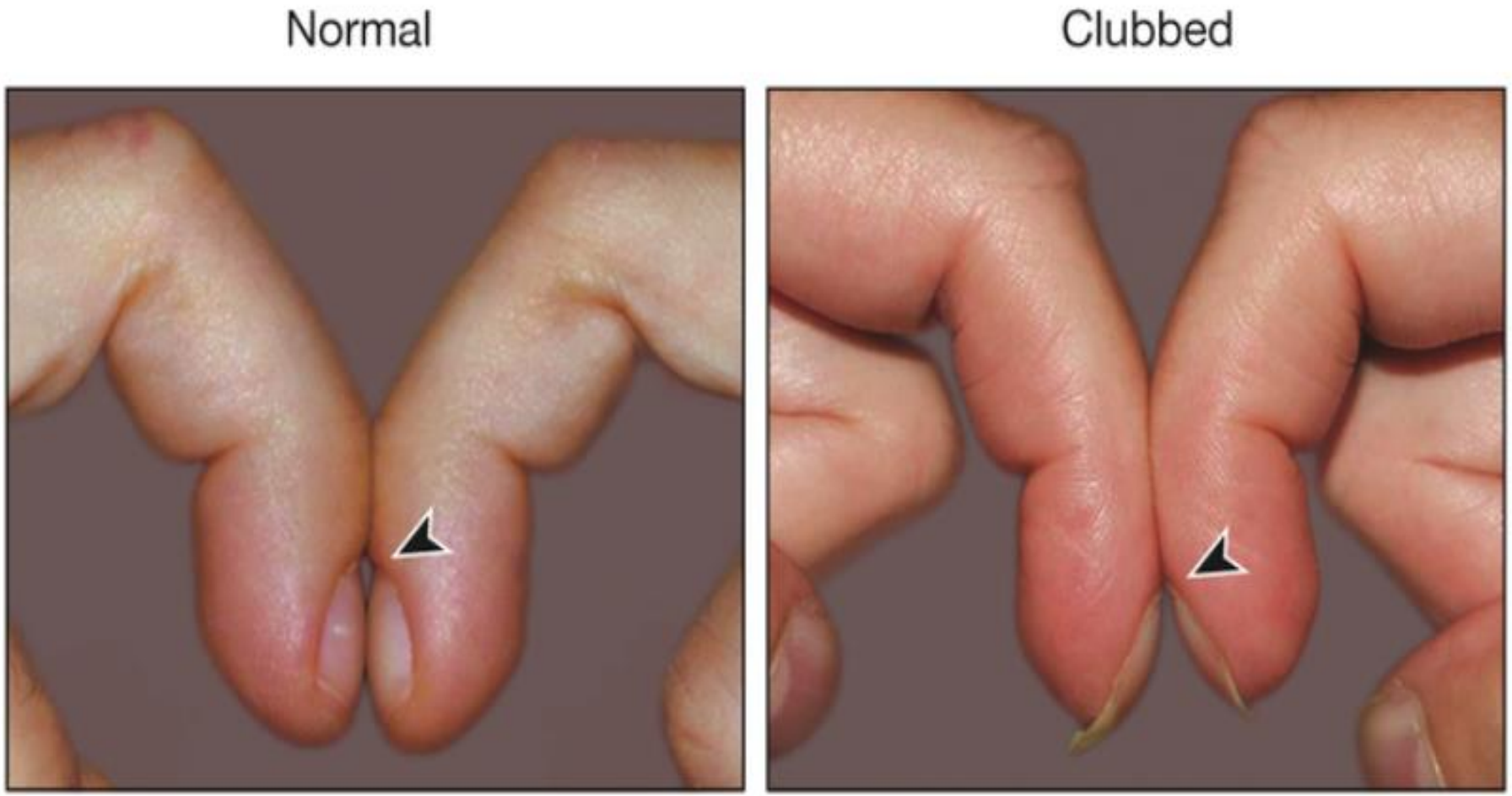

Normal
Refills within 2 seconds
Delayed
Refill takes longer
than 2 seconds

Normal fingers:

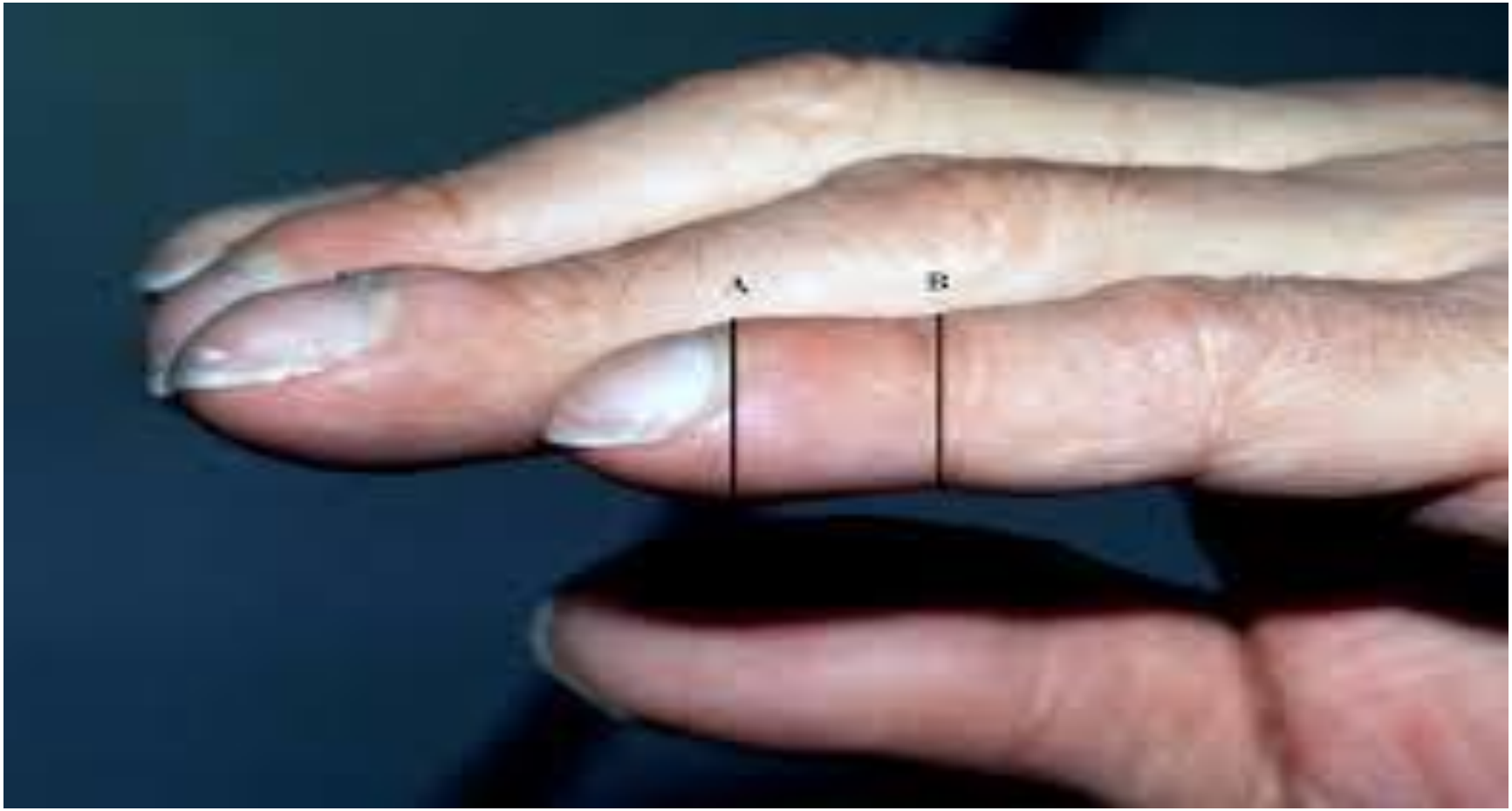



160
180+
Normal Finger
Clubbed Finger

Window test :



Parrot peak appearance:



Drum stick clubbing:



Source: Usatine RP, Smith MA, Mayeaux EJ, Chumley HS: *The Color Atlas of Family Medicine, Second Edition:* www.accessmedicine.com
Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

Hypertrophic osteodystrophy:



splinter hemorrhage



Lower limbs :

, clubbing, sandal gap in down, Marfan sign in rickets, lower limb odema, skin rash, bow knee or knock knee ,

Palpation of the dorsalis pedis and post.tibial artery

Dorsalis pedis:

Just lateral to extensor tendon of the big

toe.

post. Tibial artery:

Behind medial malleolus.

Pitting edema in foot:



Lower limb edema





Steps of the skill
Items to be fulfilled
Patients' greeting
Introduce yourself to the patient
Take the permission and explain to him or her what you are going to do
Make sure there is a good light, and that the room is warm
Proper position of the patient according to each item
Proper exposure of the patient according to each item
Maintain confidentiality and respect to the patient
Items to be used in general examination
General appearance of the patient Alert or not-Well or comfortable or not, Nutritional status (Average or under or over body built) Hydration status (Dehydrated or not)
Vital signs Heart rate -Blood pressure -Respiratory rate- Temperature

Head examination

Complexion (pallor-jaundice-cyanosis)

Shape –Facies (down, thalassemic , doll , senile , cretinism)

Congenital anomalies

Neck examination

Abnormal position (stiff-retracted-tilted)

Thyroid-Jugular venous pressure -Lymph nodes

Upper limb examination

Capillary refill

- Clubbing - Joints swellings

Hand deformities (polydactyl-Brachydactyly-simian crease-Broadening of lower ends of radius and ulna)

Lower limb examination

Joints swellings -Clubbing-Edema

Deformities (Rachitic bow legs- Marfan sign)

Lymph node examination

Cervical-epi trochlear-Axillary-Inguinal

At the end cover the exposed areas and thank him or her for cooperation